

Four Seasons Hotel Toronto Energy and Water Audit

60 Yorkville Avenue, Toronto, ON



Presented to:

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60 Yorkville Avenue,
Toronto, ON M4W 0A4

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1. EXECUTIVE SUMMARY

Four Seasons Hotel Toronto retained Morrison Hershfield Limited to prepare an Energy and Water Audit for the building located at 60 Yorkville Avenue, Toronto, ON. This energy audit report provides a description of the building's energy related systems, analyzes historical energy use, benchmarks usage against similar buildings, provides an energy use breakdown, and identifies potential energy and water efficiency measures.

The purpose of this report is to identify cost-effective measures to reduce the energy and water use and associated operating costs of the facility.

1.1 Building Overview

The Four Seasons Hotel and Residences Toronto is a 54-story residential condominium and luxury hotel tower located at 60 Yorkville Avenue. Hotel areas include P1, P2(partial) and ground floor to 21st floor. The hotel opened on October 5, 2012. The hotel building has an estimated gross floor area of 29,909 m² (321,944 ft²) and average guest occupancy~190 per day. The hotel's floor space extends from parking level 2 (P2) to the middle mechanical room, with the exception of the residential lobbies and vestibule on the building's southeast side. The hotel features meeting spaces, banquet halls, restaurants, pools, fitness rooms, spa and yoga rooms, and 259 guest rooms.

The typical hours of operation for the building are 24/7, with the exception of office areas, meeting rooms, banquet halls, and restaurants.

There are pot lights, MR 16s, aluminum reflectors with a semi-circle, and T-bar ceiling luminaires throughout the building, the majority of which were LED fixtures, except for a few unique fixtures found in the external area and guest rooms. In the public area, it is expected that 95% of the light fixtures have already been upgraded to LED models. It was reported approximately 60% of the guest rooms are equipped with LED fixtures and the remaining guest rooms light fixtures will be retrofitted to LEDs by end of 2023.

The primary HVAC systems utilize hydronic heating and cooling, with two chillers on parking level 2 (P2) and four gas-fired hot water boilers on the mechanical penthouse supplying chilled and hot water to the fan coil units, AHUs, and MAUs through hydronic pipes. The evaporative cooling towers are located on the rooftop. Two steam boilers (one gas-fired and one electric) are located on the P3 and provide steam for the laundry and humidification to the AHUs ACS-1- 6.

1.2 Energy and Water Use

From January 2019 to December 2019, the facility consumed an estimated 37,794 GJ of electricity, 45,261 GJ of natural gas and 81,953 m³ of water at a total cost of \$1,854,779 (\$31.10 per m² GFA). The split of utility costs is shown in Figure 1.

The facility has an energy use intensity of 2.78 GJ/m² GFA (498 ekWh/ m²). The facility has a water use intensity of 316 m³/per guest room per year.

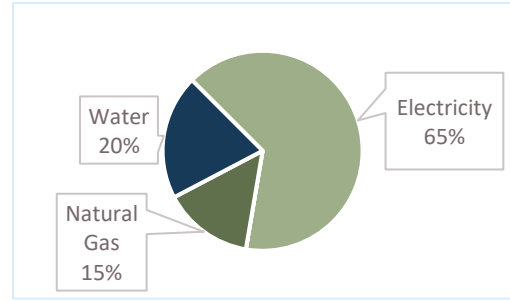


Figure 1: Utility Cost Proportions

Table 1 below summarizes the energy and water intensities for the current facility, benchmark, and target intensities based on the proposed measures. It is recommended that the facility apply energy and water efficiency measures and continue with periodic energy and water audits at least until the targets are reached. Targets should also be periodically re-evaluated.

It should be noted that although the building is currently being operated at optimal conditions, the energy and water intensities for this facility are higher compared to benchmarks due to the unique amenities such as pools, steam rooms, saunas, spas, ballrooms, conference rooms, meeting rooms, snow melt systems in the parking garage and commercial kitchens. In order to better reflect the energy consumption, we have subtracted the available sub metered data of some of the luxury items such as the kitchen, laundry, and snow melt systems and is reflected under the “altered” data column in Table 1.

Table 1: Energy and Water Benchmarks and Target

Utility	Current Facility	Current Facility Altered	Benchmark	Target
Energy (GJ/m ²)	2.78	2.13	0.88	2.00 (excluding luxury amenities)
Water (m ³ /per guest room per year)	316	-	289	299

A number of measures have been recommended and are ranked in Table 2. Further details of each measure are provided in Table 9 under the Energy Efficiency Measures section of this report. The energy audit shows that an annual utility cost savings of \$156,207 representing a savings of 8.4% in annual utility cost, are achievable. Additionally, a reduction of 459,645 kg of eCO₂, representing 18.1% of annual greenhouse gas emissions, are achievable.

Table 2: Prioritized List of Measures.

ID#	Description	Annual Savings							Initial Costs and Payback		
		Electricity (kWh)	Demand (kW)	Natural Gas (m³)	Water (m³)	Cost (\$/year)	Maintenance (\$/year)	GHG Emissions (Kg CO2)	Opinion of Cost (\$) *	Simple Payback (years)	Simple ROI
5.3.1	Replace Lavatory Aerators with Low Flow	0	0	4,739	1,040	\$5,844	\$0	9,104	\$42,000	7	14%
5.3.2	Replace Shower Heads with Low Flow	0	0	18,881	3,468	\$20,189		36,271	\$34,000	2	59%
5.4.1	Replace Urinals Flush Valves	0	0	0	34	\$158	\$0	-	\$19,000	120	1%
5.4.2	Replace Water Closets with Ultra Low Flow	0	0	0	328	\$338	\$0	-	\$396,000	-	0%
5.4.3	Install Occupancy Sensors in the Stairwells T8 Fixtures	14,418	0	0	0	\$1,649	\$429	367	\$8,600	4	24%
5.4.4	Install Occupancy Sensors in the Fitness Room	8,854	0	0	0	\$1,013	\$0	225	\$1,100	1	92%
5.5.1	Replace Hot water boilers with Condensing Boiler at the end of useful service life	0	0	76,288	0	\$95,708	\$0	150,573	\$1,502,000	16	6%
5.5.2	Install HRV units on ACS 1, 4, 5, 6 AHUs	0	0	60,885	0	\$14,299	\$0	120,171	\$100,000 **	6	18%
5.5.3	Install HRV System with Kitchen Ecology Units KEU-1,2,3,4&5	0	0	72,418	0	\$17,008	\$0	142,933	\$95,000 **	5	20%
		23,272	-	233,212	4,871	\$ 156,207	429	459,645	2,197,000	14	7%

* Net investment includes opinions of probable cost (including tax) and estimated available incentives if applicable (incentives are one-time payments, contingent on third party approval; programs may change without notice).

**Incremental Costs



2. INTRODUCTION

2.1 Terms of Reference

Four Seasons Hotel, Toronto retained Morrison Hershfield Limited to prepare an Energy and Water Audit for the building located at 60 Yorkville Avenue, Toronto, ON. Authorization to proceed with this assignment was provided by Jiri Samal, Director of Engineering on November 8th, 2022.

2.2 Objectives and Methodology

The purpose of the Energy and Water Audit is to identify energy and water savings opportunities that improve energy efficiency, and reduce operational and maintenance costs, while maintaining or improving occupant comfort indoor air quality and building performance.

The audit provides an overview of energy related systems, analyzes historical utility use, benchmarks usage against similar buildings and provides an energy end-use breakdown.

Morrison Hershfield prepared this report based on site review, document review and interviews with building operators and managers. The audit is consistent with an ASHRAE level 2 Energy Survey and Analysis, including water systems.

The facility walkthrough was conducted on December 15, 2022.

2.3 Project Team

The personnel from Morrison Hershfield involved with the preparation of this report and their responsibilities are listed below.

- Marlon Katukurunda, Building Energy Consultant, was the project manager and technical lead overseeing delivery of the report; and
- Md Maniruzzaman Akan, MAsC., P.Eng., CEM., Building Energy Consultant, conducted site review, utility analysis, analyzed energy efficiency measures, and contributed to the report.

Four Seasons Hotel, Toronto personnel that kindly provided project and on-site support include:

- Jiri Samal, Director of Engineering.

2.4 Documentation Reviewed

The following documentation was reviewed in the preparation of this report:

- Building automation system as-built drawings and operation and maintenance information package dated August 10, 2010;
- Architectural drawing dated August 1, 2007, by Architects Alliance;
- Mechanical drawing dated August 15, 2008, by MMM Group;
- Electrical drawing dated August 1, 2007, by MMM Group;
- Lighting Plan- Electrical dated August 1, 2007, by MMM Group;
- FTY- Meter Reading 2017 excel spread sheet by Four Seasons Hotel, Toronto
- Utility 2022 Budget excel spread sheet by Four Seasons Hotel, Toronto.

2.5 Limitations

This report has been based on a visual review of the facility, a review of available documents and interviews with facility staff. No tests, design review, quantity surveys, or instrument testing was carried out, unless specifically mentioned in the report. Consequently, recommendations may require change if further investigation or additional testing indicates that the initial conclusions require adjustment.

It is a basic assumption that any correspondence, material, data, evaluations and reports furnished by others are free of latent deficiencies or inaccuracies, except for apparent variances discovered during the completion of this report.

Capital costs are preliminary opinions based on appropriate costing references and must be verified by competitive tendering based on further detailed engineering efforts. Utility incentives are estimated for information purposes only. Incentives are contingent on third party approval and may change without notice.

In issuing this report, MH does not assume any of the duties or liabilities of the designers, builders or owners of the subject property. Owners, prospective purchasers, tenants or others who use or rely on the contents of this report do so with the understanding as to the limitations of the documents reviewed, the general review undertaken and understand that MH cannot be held liable for damages they may suffer in respect to the purchase, ownership, or use of the subject property.

3. FACILITY DESCRIPTION

The Four Seasons Hotel and Residences Toronto is a 54-story residential condominium and luxury hotel tower including one mid-mechanical level and one upper-mechanical level in the Yorkville district of Toronto, Ontario. Hotel areas include P1, P2(partial) and ground floor to 21st floor. The hotel building has an estimated gross floor area of 29,909 m²(321,944 ft²) and received 190 guests on average each day in 2019 based on the data provided. The hotel's floor space extends from parking level 2 (P2) to the middle mechanical room, except for the residential lobbies and vestibule on the building's southeast side. The hotel features meeting spaces, banquet halls, restaurants, pools, fitness rooms, spa and yoga rooms, and 259 guest rooms.

The typical hours of operation for the building are 24/7, with the exception of office areas, meeting rooms, ballroom, banquet halls, and restaurants. The epidemic has, however, decreased the hotel's operational capacity.



Figure 2 : Four Seasons Hotel Toronto, Main Entrance

3.1 Building Envelope

3.1.1 Exterior Walls

Generally, exterior wall material types are consistent through all the building sections. Hotel tower consist of curtain walls and vision panels.

3.1.2 Fenestration

The three main entrance vestibules consist of two sets of thermally broken double glazed, aluminum framed doors at the exterior. At the interior, there are four aluminum framed doors used to create vestibule.

The building includes aluminum punched windows and double-glazed spandrel glass aluminum curtainwall systems. Glazing consists of tinted, argon filled insulating glazing units. Some IGUs include a laminated outer pane, while the remainder have a low-emissivity coating on surface number 2. Where observed, glazing was original.

3.1.3 Roofing

It is unclear from the drawings provided the construction of the roofing system. However, it is assumed roofing consists of a protected membrane roofing system.

3.2 Electrical Distribution

Electricity is provided to the building via two main Toronto Hydro vaults. The main building is supplied by two 4,000 A bus ducts assumed to be located in level P1.

Back-up emergency power is provided to the building by one 2,080 kW standby diesel generators, located in the upper mechanical room. The generators are tested weekly by the hotel maintenance staff and semi-annually/annually by Cumming Canada.

3.3 Lighting Systems

Refer to APPENDIX B: Lighting List for a survey of lighting fixtures.

3.3.1 Office

The office spaces are typically lit by two-lamp 18W T8 LED luminaires strips. It is assumed that approximately 10% of fixtures remain on after hours for egress. The open office space lighting is controlled on a manual switch. Usually, lights are to be turned on from 8:30 am to 5:00 pm, Monday to Friday and whenever required. Meeting rooms and boardroom lighting is controlled by manual switch. The security rooms are assumed to be scheduled 24/7.

3.3.2 Corridors and Elevator Lobbies

The lighting in circulation spaces generally consists of surface/wall-mounted 2' x 4' two-lamp 48" LED luminaires and 32 W T-Bar ceiling-mounted LED luminaires. Circulation lighting is assumed to be scheduled 24/7.

The lighting in circulation spaces generally consists of surface/wall-mounted LED luminaires. Circulation lighting is assumed to be scheduled 24/7.

3.3.3 Washrooms

There are several washrooms on the P2, P1, 2, 3, 4, 6, and 9th floors of the building, including barrier-free washrooms for men and women. The restrooms are 24/7 and controlled by motion sensors.

3.3.4 Stairwell

The lighting in stairwell generally consists of surface/wall-mounted LED luminaires. They are assumed to be on 24/7

3.3.5 Guest Rooms

During the site assessment, a sample illumination assessment was done in rooms 1218 and 405, respectively. The majority of the lights are plug-ins, with the exception of a few ceiling-mounted light fixtures. Approximately 60% of light fixtures have already been replaced with LED luminaires. The complete conversion to LED is in progress and will be finished by the end of 2023. The lights are expected to be on for around eight hours for occupied rooms each day.

3.3.6 Ball Rooms

The lighting in the ballroom kitchen on the 3rd floor generally consists of 18W LED luminaire strips that is controlled by smart lighting management system. The Lutron Cove smart lighting management system has a dimming option to meet the illumination requirement of the occupants.

3.3.7 Chiller Room

LED strips typically provide lighting in the chiller room on the P2 floor. Two lighting circuits make up the system. One circuit is on a 24/7 emergency circuit. The other circuit uses motion sensors to control the lighting.

3.3.8 Fitness Room

The lighting in the fitness room on the 8th floor is provided by 18W LED luminaire strips. The lights are controlled and schedules using the Lutron smart lighting management system.

3.3.9 Garbage Rooms

Typically, two-lamp 32W 48" industrial LEDs are used for lighting in the garbage room on the P2 floor. A manual switch controls the garbage rooms' lighting. Usually, lights are to be turned on from 6:00 am to 6:00 pm, every day and whenever requires.

3.3.10 Pantry Rooms

Typically, 18W LED strips are used for lighting in the pantry rooms on the ground floor and 3rd floor. The fixtures are equipped with motion sensors.

3.3.11 Pool Area

The lighting in the pool area on the 9th floor consists of 18W LED luminaire strips. The lights are controlled and schedules using the Lutron smart lighting management system.

3.3.12 Kitchen

Typically, LED luminaire are used for lighting in the kitchen and kitchen preparation area on the 2nd floor and 6th floor. A manual switch controls the kitchen rooms' lighting. Usually, lights are on 24/7.

3.3.13 Lounge area

The lighting in the lounge and bar pool area on the ground floor consists of 18W LED luminaries strips. The lights are controlled and schedules using the Lutron smart lighting management system.

3.3.14 Mechanical Rooms

Typically, LED luminaires are used for lighting in the mechanical rooms on the P3, P1, 2nd, 4th, 5th, 7th, 8th, and mid-mechanical rooms. The system consists of two circuits one is on emergency which is on 24/7, and another circuit is installed with a manual switch that controls the mechanical room lighting, which is assumed to run 12 hours a day.

3.3.15 Storage Rooms

There are several storage rooms on most floor levels of the building. All storage rooms use LED luminaires with two circuits. Lights fixtures are on emergency circuits are on 24/7, and another circuit is installed with motion sensors which are assumed to run 12 hours a day.

3.3.16 Telecom and IT Rooms

There are several telecoms, and IT rooms on most building floor levels. All telecoms and IT rooms use LED luminaires with two circuits. Light fixtures are on emergency circuits are on 24/7, and another circuit is installed with motion sensors which are assumed to run 12 hours a day.

3.3.17 Exit Signs

The exit signs are lit by LED light sources.

3.3.18 Building Exterior

Wall-mounted down lighting fixtures that illuminate the front entrance and perimeter illuminate the outside lighting. The astronomical clock is used to control the lights using Lutron smart lighting management system.

3.4 Central Cooling Systems

There are two water-cooled chillers of capacity 2,286 kW on the P2 level and two induced draft counterflow EVAPCO cooling towers located on the roof of the building that serves the hydronic cooling system of the building's guest rooms and other spaces. The building is equipped with a free cooling system with an outdoor air reset mechanism that switches from mechanical cooling system to free cooling once the sensor read 8°C outdoor air temperature and below for 6 hours.

3.4.1 Water Side Free Cooling

The cooling plant is also equipped with a water side economizer to provide free cooling when conditions are favorable. The Super Changer plate and frame heat exchangers HEX 12 and HEX 13 are installed across chillers CH 1 and CH2 to cool chilled water directly with condenser water.

Condenser water is circulated through the heat exchanger via condenser pumps, while chilled water is circulated by the chilled water secondary pumps.

Water side free cooling is enabled when the cooling plant is in winter mode.

3.5 Central Heating Systems

The building is equipped with four (4) Unilux boilers natural gas-fired hot water boilers located in the penthouse mechanical room on the 52-floor that serves the primary heating loop. These boilers are shared between the hotel and residences. However, the portion of the hot water being supplied to the hotel primary heating loop is metered using a BTU meter.

The primary hot water loop serves several secondary systems via plate and frame heat exchangers, which supply hot water to the pool area, domestic hot water system, laundry, AHUs, fan coil units, snow melts, and glycol system, and etc.

3.6 Fan Systems

Refer to APPENDIX A: Equipment List for a list of air handling equipment and fans.

3.6.1 AHUs Serving Different Spaces of The Hotel

The main ventilation systems for the building consists of seventeen variable air volume (VAV) air handling systems located on the fan rooms (ACS-1,2,3; P2 level, ACS-4,5,6; 2nd floor; ACS-7,8,9,10,11,12 13; 7th floor, and ACS-14,15,16 and 17 on mid-mechanical room).

Each air handling unit system serves the area shown on appendix A. The AHUs are equipped with hydronic heating coils and cooling coils and gas-fired/electric humidifiers which shown in Appendix A.

AHU ACS-1, 2, and 3, which are situated on parking level 2, provide service to the BOH P1 and P2 in the basement as well as the bar and restaurant on levels L1 and L2.

The second floor's ACS is called ACS-4, 5, and 6, and it serves the hotel and residential lobby on the ground floor and the level 2 and level 3 administrative rooms.

The L3 Ballrooms and preparation area, L6 Banquet room, L6 pre-function & meeting rooms, L8 workout, L9 Spa, and L9 pool area are served by AHUs identified by ACS-7, ACS-9, ACS-10, ACS-11, ACS-12, and ACS-13, which are located on the 7th level.

ACS-13 serves the pool, which has a sensible efficiency of 90%, ACS-7, 9, and 12 use heat recovery wheels with a sensible effectiveness of 58%. All AHUs are controlled by the building automation system.

The following control strategies are typically used by BAS for this system:

- The system is scheduled to run as per set up control strategy on BAS.
- The heating and cooling coil maintained 13°C discharge air temperature.
- Enthalpy sensors in OA and RA to reduce outdoor air to minimum setting when outdoor enthalpy rises above indoor enthalpy.
- Pressure sensor to modulate fan speed to maintain constant pressure in supply duct system, located $\frac{3}{4}$ of the way down the supply air main.
- Indoor outdoor pressure sensors controlling recirculation exhaust fan speed to maintain slight positive pressure in the building.
- Minimum outdoor air damper set point to be 30% or as identified on AHU schedule adjusted through BAS.
- Humidity in RA duct will control steam humidifier and cooling coil to maintain 30% RH in winter and 45% RH in the summer adjustable by Bas.

- Low temperature sensors enable and control through BAS to protect freeze state.

Provide with high humidity sensors ensure RH not to exceed 90%.

3.6.2 MAUs System - MAU1 to MAU7

A total of seven MAUs are available for the airlock, retail, main kitchen, ballroom kitchen, banquet kitchen prep, kitchen on the sixth level, and hotel corridors.

MUA 6 and MUA 7 is equipped with a heat recovery wheel with a sensible effectiveness 58% that extracts heat from the hotel washroom exhaust.

All MUA units are equipped with a glycol heating coil and supply fan. The heating coil is equipped with two coil circulators.

The BAS implements the following control strategies for this system.

- The system is scheduled to run continuously.
- Heating coil capacity control is by provided by modulating glycol temperature via steam to glycol heat exchanger.
- The discharge air temperature has a constant setpoint of 70°F.

3.6.3 Swimming Pool ACS-13 (Different Construction for Pool Use)

ACS-13 is an Ingenia CAH-2SP unit located in the Fitness room. The unit is controlled and maintains the required pool environment using an algorithm provided by Johnson Controls with this unit.

3.6.4 Variable Air Volume Boxes

The building is equipped with VAV boxes. All VAV boxes are digital, pressure independent models.

VAV boxes modulated from minimum and maximum position on a call for cooling to maintain space temperature setpoints.

3.6.5 Fan Coil Units (FCUs)

Each guest rooms, vestibules, and 3rd floor corridor are equipped with fan coil units (FCU), which circulates air, while hydronic heating or cooling coils offer heat in the winter and cool air in the summer. The Inncom guest management system is used to control the FCUs availability based on the guest check-in and presence in the rooms.

3.6.6 Unit Heaters

The building is equipped with hydronic unit heaters in the parking area, loading area, boiler room, mechanical rooms, and water service fire protection rooms. The heaters are operated on 50% propylene glycol and maintain delta of ~16°C. The leaving temperature of the air is maintained at ~40°C.

3.6.7 Humidifiers

The building is equipped with one Unilux natural gas-fired steam boiler and one Bryan electric steam boiler located on the P3 level provides humidification for ACS1,2,3,4,5 and 6. The boilers have 80% (Bryan) and 95% (Unilux) thermal efficiency, respectively.

The building is also equipped with gas steam humidifiers serving MUA 6 and 7, and individual electric steam humidifiers serving ACS-7-12, and ACS-14-17.

3.6.8 Exhaust Systems

The building is equipped with a number of exhaust fans providing parking, washroom, kitchen and general exhaust. The fans are located throughout the building; however, a number of fans are located in the 2nd, 5th, 7th, 10th(roof) mid mech, penthouse, and other locations of the building. Most of the exhaust fans are controlled by the building automation system and run on a time-of-day schedule, although most fans appear to run 24/7

3.6.9 AC Units

There are several water-cooled AC units connected to the condenser water loop that serves the main server room, telecom room and other area of the building. All AC units are being operated continuously to provide cooling.

3.7 Domestic Water & Plumbing Systems

Potable water is delivered from the City of Toronto service to the building through the main connections at the water entry room on the P1 level mechanical room and the water meter located on the P2 level. City water enters the building and splits, feeding fire protection systems and potable water systems, where it is metered before distribution.

Refer to APPENDIX A: Equipment List for a list of DHW related equipment.

3.7.1 Domestic Hot Water

Domestic hot water used for the plumbing fixtures is generated by hot water boilers from the central heating plant through the plate and frame heat

exchangers designated by 7A and 7B located on P3 in the P3 level Steam Boiler Room. The system consists of one hot water storage tank, a circulating pump, and a condensate return unit. The BAS setpoint of 145°F for this hot water system. Domestic cold water is supplied for the generation of domestic hot water for the plumbing fixtures and the water softener conditions other domestic hot water uses.

3.7.2 Kitchen and Laundry Hot Water

Domestic hot water used for the kitchen and laundry is generated by hot water boilers from the central heating plant through the plate and frame heat exchangers designated by 2A and 2B located on P3 Steam Boiler Room. The system consists of one hot water storage tank, a circulating pump, and a condensate return unit. The BAS setpoint of 145°F for this hot water system.

3.7.3 Pool and Whirlpools

An uninsulated Armstrong, shell and tube heat exchanger located on the 9th floor serves the pool heating. The system has a hot water temperature at the pool setpoint of 84°F on the BAS. Hot water is circulated via pumps.

The whirlpool is heated by a shell and tube heat exchanger, located in the same floor. The whirlpool water is circulated by a pump. The whirlpool has a setpoint of 104°F.

3.7.4 Ice Machines

Each floor is equipped with a Manitowoc ice cube machine providing ice serving guest rooms. The various kitchens are equipped with additional Manitowoc ice machines. The units are equipped with water cooled condensers that reject heat directly to the space.

3.7.5 Plumbing Fixtures

Plumbing fixtures for the building are installed in public washrooms, changerooms, kitchen/cafeteria, kitchenettes, janitor's closets, amenity spaces and in the suites.

information based on the site survey

Table 3 contains the plumbing fixture information based on the site survey

Table 3: Plumbing Fixture Summary.

Group	Fixture	Flush/Flow Rate	Approximate Quantity
Public and staff	Water Closet	13.5 Lpf	60
	Urinal	3.8Lpf	15

Group	Fixture	Flush/Flow Rate	Approximate Quantity
	Lavatory	5.7 Lpm	49
	Shower	9.5 Lpm	17
Guest Room	Water Closet	6 Lpf	253
	Lavatory	7 Lpm	560
	Shower	9.5 Lpm	284

3.8 Telecom and Server Room Loads

The facility is equipped with telecom and server rooms with loads ranging from 30kVA to 80 kVA. It is assumed that data loads operating at 40% capacity.

Refer to APPENDIX A: Equipment List for a list of telecom and server room equipment.

3.9 Building Automation and Controls

The facility is equipped with Direct Digital Control (DDC) system provided and maintained by Johnson Control Metasys system. The central Building Automation System (BAS) provides control over and/or status of air handling equipment, fans, pumps, boilers, and domestic hot water systems.

3.10 Vertical Transportation

The facility is equipped with fifteen (15) elevators of different sizes and capacities serving different levels of the buildings. Among those, one is a cargo elevator, and the remaining others are passenger elevators.

3.11 Miscellaneous Equipment

Miscellaneous equipment is generally comprised of the following:

- Meeting room AV systems;
- Kitchenette and coffee station appliances;
- Fitness equipment;
- Security systems; and
- Communications and IT systems;
- Steam room equipment;
- Spa room equipment;

4. UTILITY ANALYSIS

The building is connected to the Toronto Hydro power grid. Natural gas is delivered by Enbridge Gas distribution network. Potable water is provided to the building via the City of Toronto's potable water supply. Each service is metered by the respective utility company.

An excel utility budget spreadsheet was obtained that included utility data from January 2013 to September 2022 for electricity, natural gas and potable water. Electricity falls under Toronto Hydro 50 to 99 kW general service while natural gas falls under Enbridge Commercial and Industrial Rate 6.

4.1 Baseline Year

A baseline year was established using the utility data from January 2019 to December 2019. Calendarized utility data for each nominal month is in APPENDIX C: Utility Bill Records. A summary of the baseline year's utility usage is provided in Table 4 : Utility Summary for Baseline Year below for the site.

Table 4 : Utility Summary for Baseline Year

Utility	Consumption	Peak Demand	Cost	Blended Rate
Electricity	10,498,471 kWh (37,794 GJ)	-	\$1,209,088	11.52 ¢/kWh (\$32.00 /GJ)
Natural Gas	1,154,154.58 m ³ (45,261 GJ)	-	\$271,064	23.49 ¢/m ³ (\$5.99 /GJ)
Water & Sewage	81,953 m ³	-	\$374,627	\$4.57 /m ³
Total			\$1,854,779	

During the baseline year, facility incurred \$1,854,779 in utility costs. Approximately 65.2% of the total utility costs were due to electricity, 14.6% from natural gas, and 20.2% due to water. The blended utility rates above represent the total utility cost, which generally include consumption, demand, and fixed charges, divided by the consumption. These blended rates above are used for calculating cost savings associated energy efficiency measures.

Energy and water consumption for the baseline year are summarized in Figure 3. Electricity use peaks in July and August, whereas natural gas use coincides with colder weather in January and February. Water use shows peaks in July and August, likely due to the cooling tower makeup water consumption.

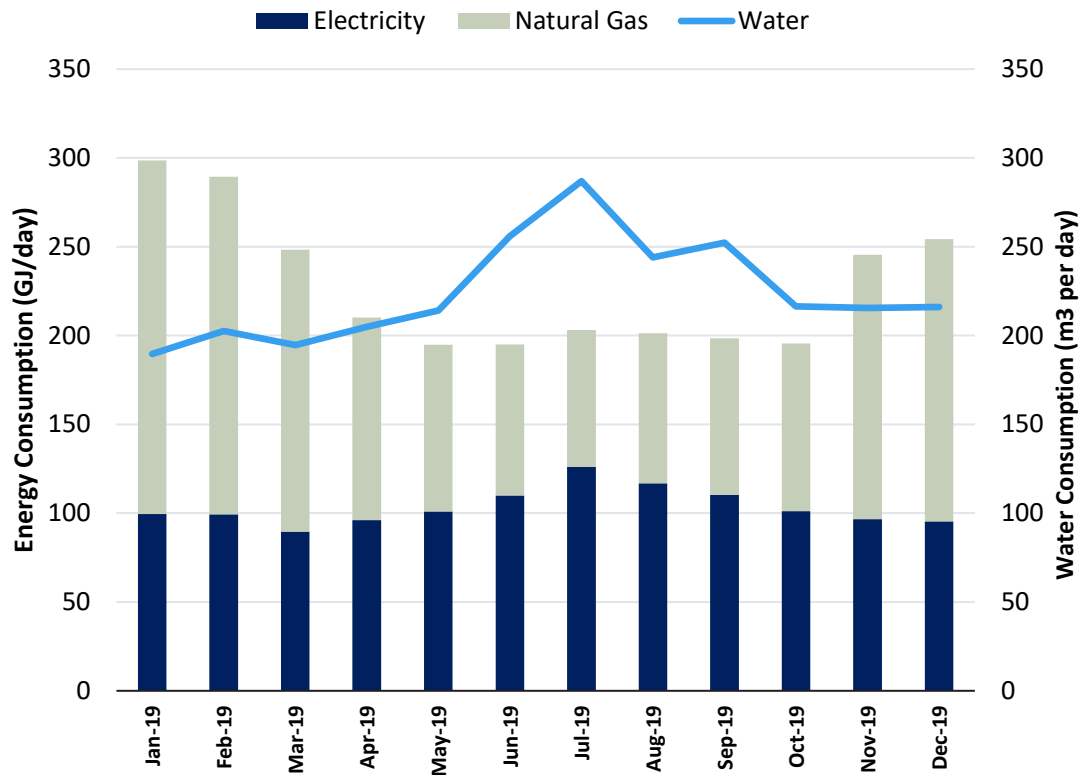


Figure 3: Baseline Average Daily Energy and Water Consumption

4.2 Historical Use

4.2.1 Electricity

The electrical load includes lighting, cooling, fans, pumps, plug loads, kitchen equipment, server/telecom room, and elevator. The average daily electricity consumption is shown in Figure 4 below from January 2017 to December 2021. Electricity consumption has larger peaks in summer and smaller peaks in winter. This is likely driven by higher cooling and fan energy in the summer, and pumps / auxiliary electric heat in the winter.

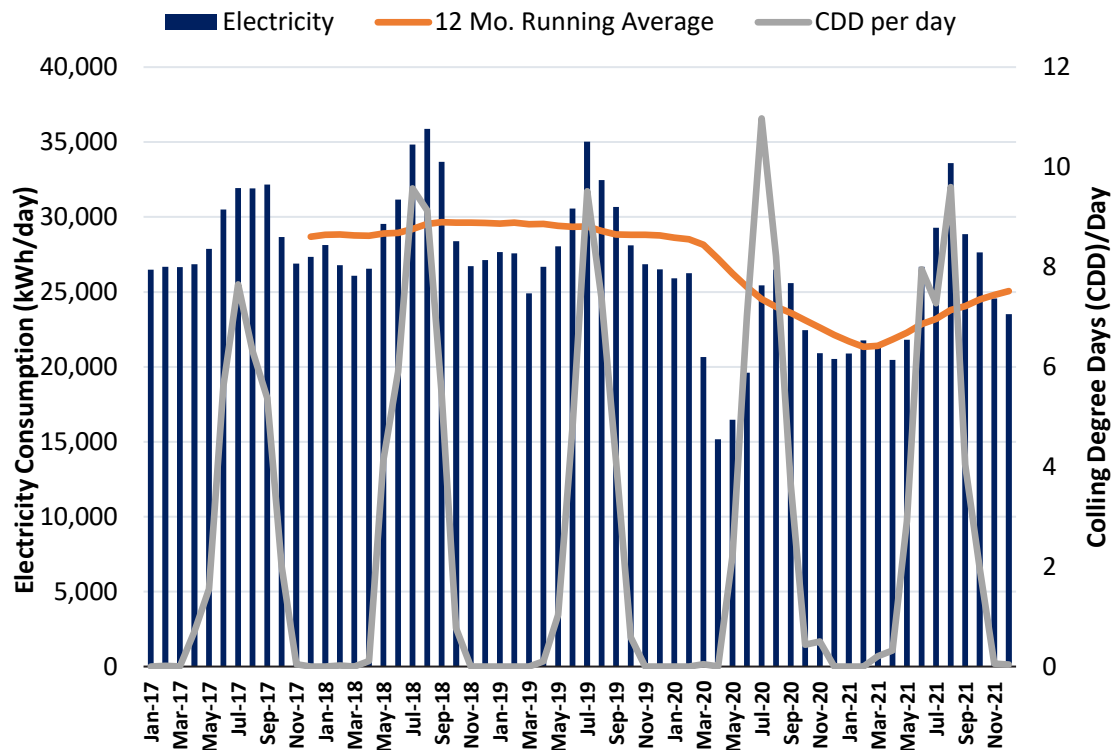


Figure 4: Average Daily Consumption of Electricity

Weather analysis of the building's electricity use showed a balance point of approximately 13°C and a moderate correlation with cooling degree days (CDD) (base 13°C). Above the balance point, electricity use is influenced by mechanical cooling. Further discussion is included in APPENDIX D: Weather Normalization.

4.2.2 Natural Gas

Natural gas is primarily consumed by the heating loop boilers, gas-fired air handling equipment, and kitchen equipment. Figure 5 shows the natural gas consumption from January 2017 to December 2021.

Natural gas use shows seasonal variations, due to space heating and humidification with a small load in the summer from domestic hot water heating and cafeteria cooking use.

Natural gas usage is similar in 2017 and 2021 according to the 12-month running average but has decreased by about 35% in 2020 than 2019, which may be related to the pandemic effect. However, it was found from the weather data that HDD was 19.5% lower in 2020 than in 2019. The site management staff informed us that the number of visitors was relatively low to very low, and the commercial kitchens and swimming pools were closed during pandemic time.

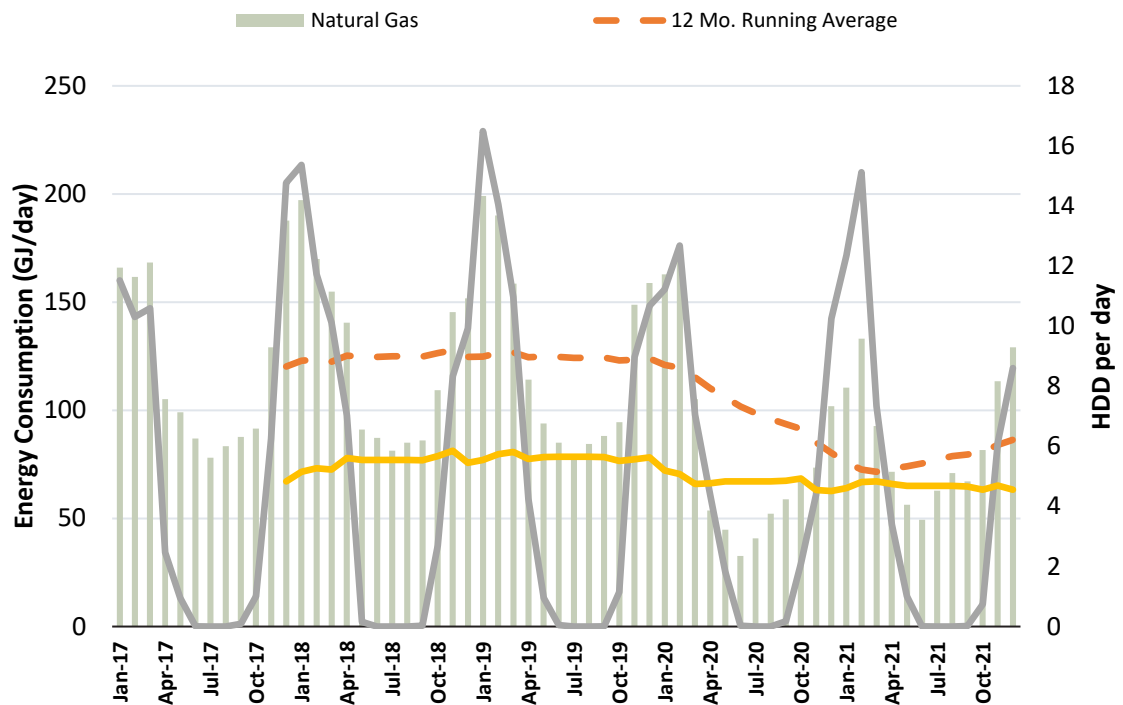


Figure 5: Average Daily Consumption of Natural Gas

Weather analysis of the building's natural gas use showed a balance point of approximately 10°C.

Above the balance point, gas use consists of domestic hot water heating and cooking while below the balance point it is dominated by heating. Weather normalization showed a strong correlation between gas consumption and heating degree days (base 10°C). Further discussion is included in APPENDIX D: Weather Normalization.

4.2.3 Water

The historical water consumption at Four Seasons Hotel Toronto from January 2017 to December 2021 is shown in Figure 6.

Water consumption peaks seasonally during the summer months due to cooling tower makeup water consumption and winter months due to humidification. The 12-month running average appears to indicate a significant downward trend in water consumption of from 2019 by 50.5 % and 40.6% in 2020 and 2021 respectively. This is likely due to the guest and full-time employee reductions in 2020 due to COVID-19.

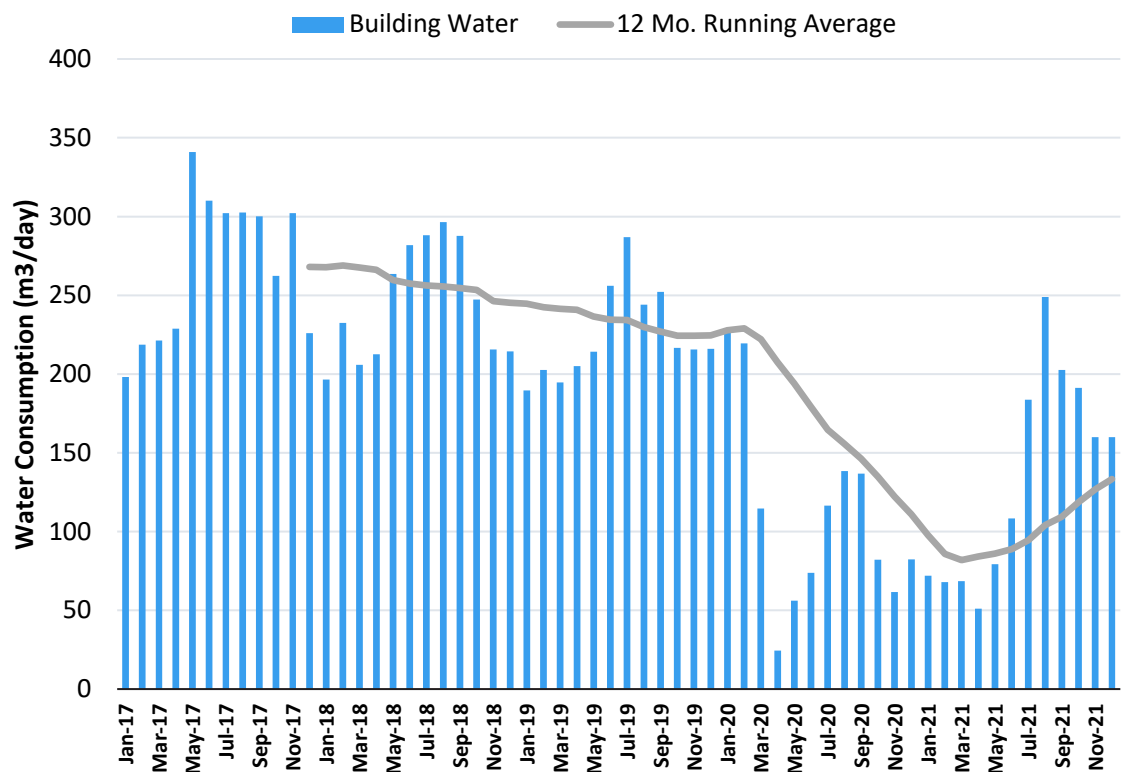


Figure 6: Average Daily Consumption of Water

4.3 Benchmarks and Targets

Benchmarking building energy use provides a standard of comparison for building performance. It can also help to set performance targets, used to formulate energy reduction strategies.

4.3.1 Energy Benchmark

The facility's energy and water use per year has been benchmarked against the Energy Star survey results for the Canadian building types in the Table 5 below. The current facility consumes 2.78 GJ/m² GFA (498 ekWh/ m²) of energy and 2.74 m³/m² GFA of water per year.

It should be noted that unique amenities provided by the hotel such as spas, saunas, steam rooms, pool, ballrooms, conference rooms, meeting rooms, commercial kitchen and other luxury amenities, impacts the energy and water use intensity of the hotel making it much higher in relation to comparable benchmarks. Although the building is operating in optimal conditions these unique amenities provided by the hotel makes it difficult to benchmark against

readily available benchmark data. In order to better reflect the energy consumption, we have subtracted the available sub metered data of some of the luxury items such as the kitchen, laundry, and snow melt systems. Energy intensities excluding these sub metered data is reflected under “Hotel Altered” data.

Sub metered data for kitchen, laundry and snow melt system was obtained from “TFY- meter reading – 2017” excel work sheet provided.

The Four Seasons hotel has been benchmarked in

Figure 7 against Energy Star survey data and 2019 Statics Canada average energy use intensities by building type.

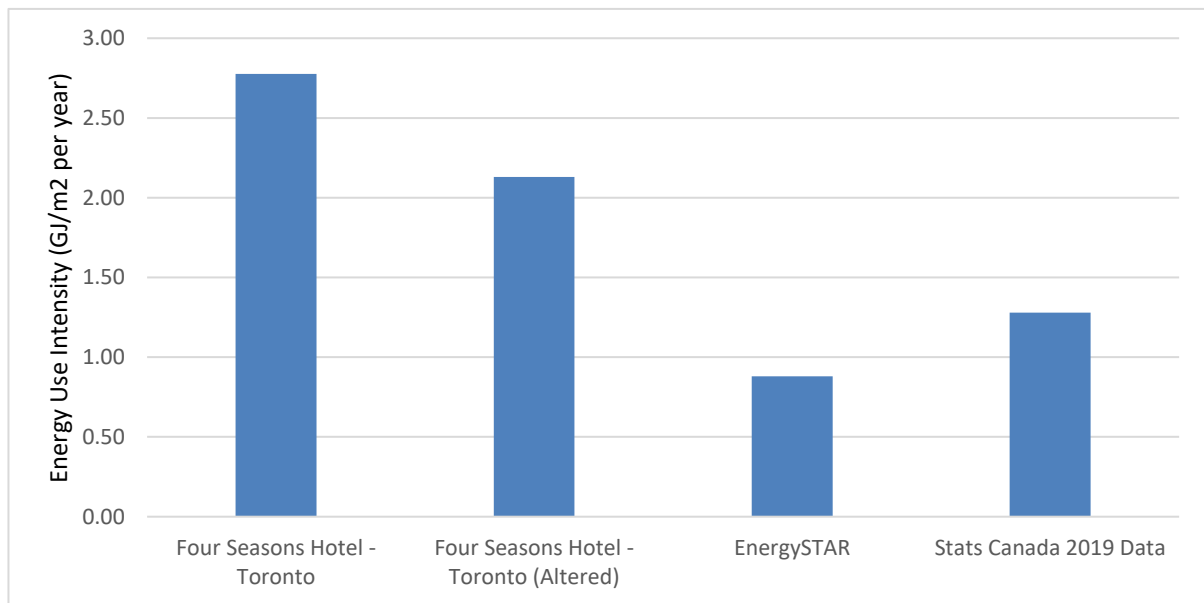


Figure 7: Energy Benchmark, Hotel Sector

The Four Seasons Hotel has a water intensity of 316 m³ per guest room per year, which is higher than the US average, as reported by the US EPA and presented in Figure 8. However, the facility’s water use is considered above average by CIRIA (Construction Industry research and Information Association, London), based on UK data for 4- and 5-star hotels with swimming pools.

Similar to the energy use, it is probable that the commercial kitchen, spa facility and the pool along with other luxury amenities that consumes water is causing the hotels water use intensity to be higher compared to the available benchmarking data.

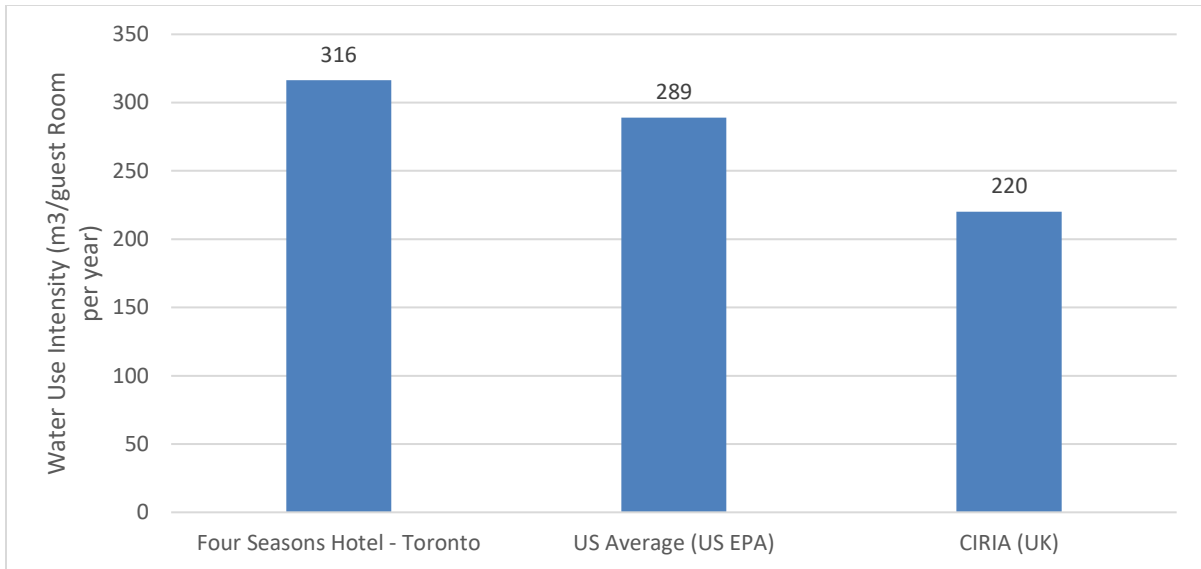


Figure 8: Water Energy Benchmark, Hotel Sector

Table 5 provides an overview of the benchmarking results against Energy Star Portfolio Managers and comparable industry averages. Water use was 9.4% greater than the US average, while energy use at the Four Seasons Hotel was 66% higher compared to Statics Canada's 2019 survey data and 142% higher compared to Energy Star benchmark data.

Table 5: Energy and Water Benchmarks

Utility	Current Facility	Benchmark EnergySTAR	Canada/US Average	% Difference than Industry Average
Energy (GJ/m²)	2.13 (Altered value)	0.88	1.28	-66%
Water (m³/guest room per year)	316	-	289	-9.4%

Setting energy and water reduction targets can help drive efficiency gains and operational cost savings. The following targets are recommended based on the energy and water conservation measures suggested.

Table 6: Target Energy and Water Intensities

Utility	Current Facility	Benchmark	Targets
Energy (GJ/m²)	2.13 (Altered value)	0.88	2.0
Water (m³/guest room per year)	324	289	299

It is recommended that the facility apply energy and water efficiency measures and continue with periodic energy and water audits at least until these targets are met. Targets should also be periodically re-evaluated.

4.4 Energy Balance

The building's energy use breakdown by major end use for the baseline year is shown in Figure 9 and presented in Table 7 below. A brief description of how each value was derived is provided below.

- Interior and exterior lighting energy was determined based on a survey of lighting fixtures in the building and typical occupied schedules and our observations of space use.
- Fan and pump energy are based on an inventory of equipment, scheduled or estimated hours of operation, and estimated motor power draw.
- Server room/ telecom closet energy use is based on the inventory of equipment and estimated average power draw of 40%.
- Plug load energy use includes receptacle loads, such as office equipment, and kitchenette equipment. Office equipment energy was calculated based on assumed power draw of equipment and hours of operation.
- Elevator energy use was calculated using the Thyssen Krupp online energy calculator.
- Cooling energy use estimated by disaggregating summer electrical peak loads from the baseload.
- The domestic hot water heating load was determined by the monthly summer natural gas use and assumed to be constant throughout the year.
- Humidification energy was estimated based the assumed hourly outdoor air load using weather data from Environment Canada for the baseline period for the electric humidifiers. Natural gas usage for gas steam humidifiers were obtained from sub metered data.
- Natural gas usage for the kitchen, laundry and steam boilers were obtained from sub metered data.
- Natural gas space heating end use was assumed to be the remainder of unaccounted gas use.

The largest energy end use is attributed to space heating, followed by lump-up of kitchen, laundry and humidification and space cooling end-uses.

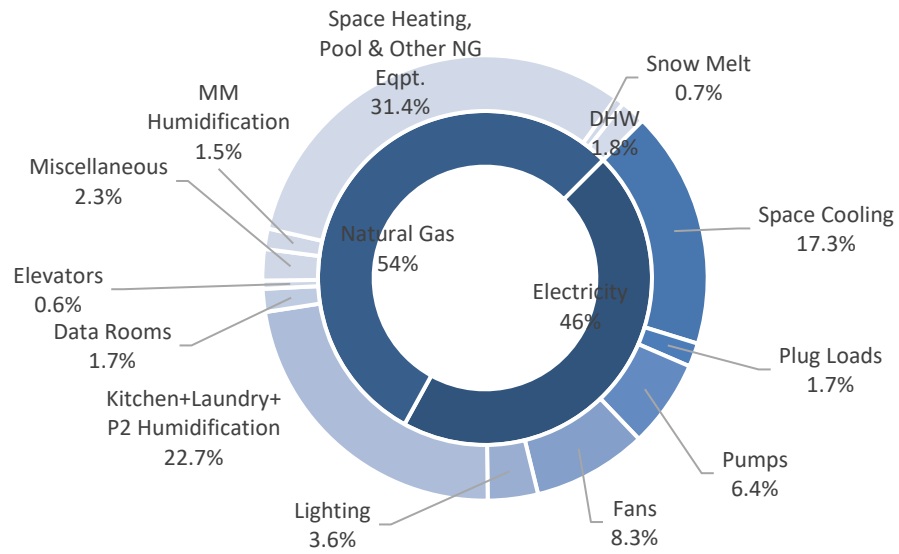


Figure 9: Energy End Use Breakdown

Table 7: Energy End Use Breakdown

End Use	Electricity (GJ)	Natural Gas (GJ)	Total (GJ)	Percent (GJ)	Cost	Percent (Cost)	GHG Emissions (Kg eCO ₂)
Space Cooling	14,339	0	14,339	17.3%	\$458,719	31%	101,301
Plug Loads	1,429	0	1,429	1.7%	\$45,729	3%	10,098
Pumps	5,291	0	5,291	6.4%	\$169,277.60	11%	37,382
Fans	6,902	0	6,902	8.3%	\$220,817	15%	48,764
Lighting	3,021	0	3,021	3.6%	\$96,655	7%	21,345
Kitchen & Laundry	2,330	16,561	18,891	22.7%	\$173,720	12%	849,989
Data Rooms	1,388	0	1,388	1.7%	\$44,390	3%	9,803
Elevators	495	0	495	0.6%	\$15,846	1%	3,499
Miscellaneous	1,890	0	1,890	2.3%	\$60,454	4%	13,350
Humidification	708	520	1,228	1.5%	\$25,775	2%	31,156
Space Heating, Pool & Other NG Eqpt.	0	26,109	26,109	31.4%	\$156,367	11%	1,314,089
Snow Melt	0	544	544	0.7%	\$3,260	0%	27,397
DHW	0	1,526	1,526	1.8%	\$9,142	1%	76,828
Total	37,794	45,261	83,055	100.0%	\$1,480,152	100%	2,545,001

4.5 Water Balance

The building's water use breakdown by major end use for the baseline year is shown in Figure 10 and presented in Table 8. A brief description of how each value was derived is provided below.

- Indoor plumbing fixtures and fittings represents the domestic cold-water consumption of plumbing fixtures, such as water closets, urinals, lavatories, showers and kitchen sinks. It is based on a survey of plumbing fixture flow rates and assumed occupant usage.
- Domestic hot water represents the hot water consumption of indoor plumbing fixtures and fittings. It is based on typical fixture supply temperatures, a survey of plumbing fixture flow rates and assumed occupant usage.
- Cooling tower water use is based on daily sub-meter readings and verified by disaggregating seasonal trends from the main water meter.

- Hotel car wash water use is based on sub-meter.
- Calculated water use for the kitchen, laundry, spa, humidification, and other uses was subtracted from the total water consumption.

The largest and most expensive water end use is attributed kitchen, laundry, pool & Spa, followed by the cooling tower makeup water.

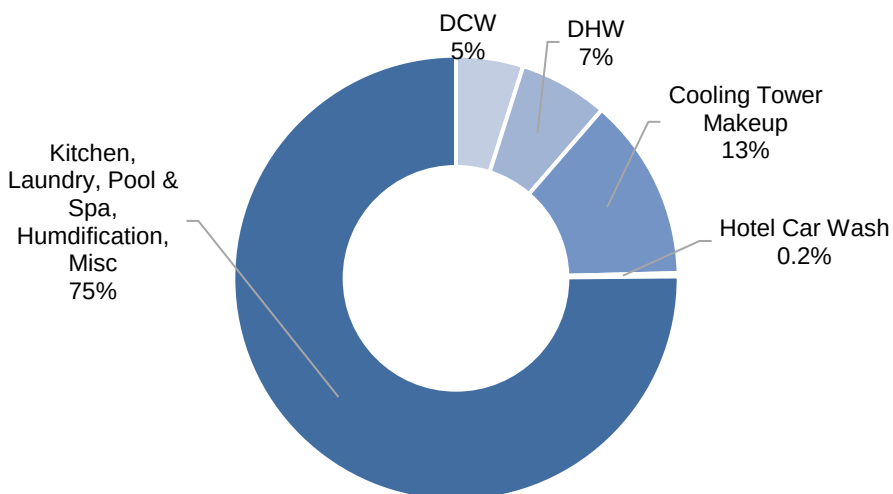


Figure 10: Water End Use Breakdown

Table 8: Water End Use Breakdown

End Use	Consumption (m ³)	Percent (m ³)	Cost
DCW	4,003	5%	\$18,300
DHW	5,304	6%	\$24,247
Cooling Tower Makeup	10,872	13%	\$49,699
Hotel Car Wash	198	0.2%	\$904
Kitchen, Laundry, Pool & Spa, Humidification, Misc.	61,576	75%	\$281,477
Total	81,953	100%	\$374,627

5. ENERGY EFFICIENCY MEASURES

This section identifies efficiency measures to improve building energy and water use, and reduce operating costs, while maintaining occupant comfort and well-being.

Several efficiency measures were identified that include policy measures, operation and maintenance measures, and energy and water efficiency measures, including low- and no-cost measures as well as potential capital upgrades. Other measures considered but dismissed have been listed as well. Measures are summarized in Table 9 at the end of this section.

5.1 Policy Measures

5.1.1 Guest Engagement

Current guest engagement strategies include signage in the guest rooms advising guests to re-use towels, which saves the water and energy that would otherwise be required for laundry.

In addition, the hotel encourages guests to turn off lights and climate control units when rooms are not in use, as advertised in the guest room concierge book and on-demand television system.

We recommend continuing with this program; however, more prominent display of recommendations to turn off lighting and climate control when not in use could increase the effectiveness of this program.

A detailed discussion has not been provided for this measure.

5.1.2 Guest Room Lighting Policy

Currently, cleaning staff leave some lighting on in guest suites after cleaning to welcome newly arriving guests. We understand that this is part of the hospitality policy; however, energy savings could be realized by revising the policy to leave lights off after cleaning.

5.1.3 Energy Star Equipment Purchasing Policies

When purchasing new equipment, consider equipment with a high Energy Star rating. This includes guest room televisions and fridges; office equipment, including monitors and printers; and commercial kitchen equipment.

Amend current purchasing policy to mandate that all new equipment be Energy Star rated, unless economically infeasible.

A detailed discussion has not been provided for this measure.

5.2 Operations and Maintenance Measures

5.2.1 Steam Trap Survey and Leak Repairs

Steam trap losses can average 25% of a facility's steam consumption per year; as such, an effective steam trap inspection and maintenance program is critical to the energy efficient operation of a steam system.

We recommend continuing with the current inspection program at least on an annual basis. Failed traps should be replaced as soon as they are identified.

5.2.2 Water Leak Inspection and Repair Program

Incorporate a vigilant leak inspection and repair program into the preventative maintenance plan, and repair water leaks immediately. In addition to reducing water usage, repairing leaks reduces the probability of secondary damage to the building structure and finishes, the risk of mould growth, and fall hazards associated with wet floors (including in mechanical service rooms).

A detailed discussion has not been provided for this measure.

5.3 No-Cost and Low-Cost Measures

Details of each No-Cost and Low-Cost Measure are provided in APPENDIX E: Energy Efficiency Measure Details, including detailed description of the proposed measures, description of the implementation, energy savings calculations and opinion of implementation costs.

5.3.1 Replace Lavatory Aerators with Low Flow

Lavatories are installed with standard-flow aerators. Replace with low-flow aerators (1.9 Lpm) to reduce water consumption and water heating energy. For best results, pair with a foaming hand soap that is easier to rinse.

5.3.2 Replace Shower Heads with Low Flow

Consider replacing shower heads with lower flow 5.7 Lpm heads.

5.3.3 Unit Heater Setpoints

Reduce unit heater setpoints in water service fire protection room, boiler rooms, loading dock, parking garage, and mechanical rooms.

We recommend reducing the temperature setpoints to 18°C (65°F) if it is above that setpoint temperature.

5.4 Capital Upgrade Measures

Details of each Capital Upgrade Measure are provided in APPENDIX E: Energy Efficiency Measure Details, including detailed description of the proposed measures, description of the implementation, energy savings calculations and opinion of implementation costs.

5.4.1 Replace Urinals

Replace the current standard flow urinals with low flow models.

5.4.2 Replace Water Closet with Ultra Low Flow

Consider replacing shower heads with lower flow 4.8 Lpm fixtures.

5.4.3 Replace Stairwells T8 Fixtures with LED Motion Sensor Luminaires

Stairwells are currently illuminated by two-lamp, T8 wrap surface mounted fluorescent luminaires that operate on a 24/7 schedule. Replacing these fixtures with LED luminaires with built in motion sensors will reduce the energy usage during peak times and dim the lighting to lower levels when not used (overnight).

5.4.4 Install Occupancy Sensors in the Fitness Room

The building has a fitness room equipped with manually operated light switches. Occupancy sensors can help minimize wasted usage when these rooms are not in use.

5.5 End of Life Measures

The following measures are recommended for further analysis at the end of service life. Energy savings alone generally will not justify the outright replacement; however, upgrades should be considered at end of life.

5.5.1 Replace Hot Water Heating Boiler with Condensing Boiler

Replace the boiler serving the hot water loop with a high efficiency condensing boiler at failure.

5.5.2 Install HRV System with ACS-1, 4, 5, 6 Air Handling Units

Install HRVs on the ACS-1, 4, 5, and 6 Air Handling Units so that heat from the stale air leaving the area can be captured and used to warm the fresh air entering the places they serve.

5.5.3 Install HRV System with Kitchen Ecology Units KEU-1, 2,3,4, &5

Healthy indoor air quality is one of the key elements of a restaurant design. The kitchen hood exhaust is the main source of ventilation load in restaurants. Heating the make-up air during the winter months entails a significant upfront cost and, more significantly, a significant ongoing expense. Heat recovery systems are always an attractive solution because of the high temperatures of hood exhaust air. There are crucial factors to consider, too, given the large upfront expenditures of heat recovery systems. An electrostatic precipitator should best protect the system with a self-cleaning automatic washing system because the exhaust air from the kitchen hood is heavily contaminated.

One of the following alternatives must be used per ASHRAE Standard 90.1 for kitchen ventilation systems with a total exhaust flow rate of more than 8,400 m³ per hour (4,948 cfm):

- 50% of the kitchen supply air must be the transfer air to be exhausted otherwise.
- A 50% efficiency heat recovery system.
- Variable speed system can control at least 75% of the exhaust flow rate. This demand-controlled system should be able to reduce the exhaust air and fresh air supply by 50%.

Install HRVs on the KUE-1, 2,3,4 & % Kitchen Ecology Units so that heat from the stale air leaving the area can be captured and used to warm the fresh air entering the places they serve. The reported costs associated with this measure is the incremental costs.

5.6 Dismissed Measures

The following measures were considered but not recommended.

5.6.1 Use Passive Indoor Space Heating by Solar Wall (Dismissed)

The solar wall passive heating system provides the greatest potential for energy savings during the spring and fall seasons when outdoor temperatures are neither too cold nor too hot. In order to heat the outdoor air using solar energy, these walls integrate outer construction with interior fixtures. The mid-mechanical room and mechanical penthouse can both have these walls renovated. Other walls can be considered to keep the aesthetic aspect of the building. Having the air preheated by the passive solar plenum reduces the amount of heating required using natural gas, thereby saving energy. A study can be conducted to identify the feasibility and cost.

5.6.2 Use District Heating and Cooling (Dismissed)

The district energy system handles the task and offers useful advantages. It typically uses less fuel and eliminates the need to install separate space heating, cooling, and hot water systems in each building when heating and cooling is provided from a central plant. Buildings with different energy profiles can be connected to a primary energy supply using a district energy system, which is a very effective design. To assess the viability of the district heating and cooling system and decarbonization elements, a study might be carried out in collaboration with Enwave, which is close to the hotel.

5.6.3 Use Daylighting in Guest Rooms and Ball Rooms (Dismissed)

The luminaires nearest to the window in the guest rooms and ballrooms were considered to be dimmed or de-energized using photocell controls during bright conditions. The high implementation costs paired with the low lighting levels on the north side of the cafeteria quickly dismissed this measure.

5.6.4 Replace Urinals with Waterless Models (Dismissed)

Due to increased maintenance costs, this option does not have a payback and is not recommended.

5.6.5 Meeting Room Demand Control (Dismissed)

When the meeting room VAV boxes reach the end of their life, investigate installing occupancy sensors to set damper position to minimum during unoccupied periods.

This measure was dismissed due to very long payback periods.

Table 9: Summary of Energy Efficiency Measures

ID#	Description	Annual Savings							Initial Costs and Payback		
		Electricity (kWh)	Demand (kW)	Natural Gas (m³)	Water (m³)	Cost (\$/year)	Maintenance (\$/year)	GHG Emissions (Kg CO2)	Opinion of Cost (\$) *	Simple Payback (years)	Simple ROI
5.3.1	Replace Lavatory Aerators with Low Flow	0	0	4,739	1,040	\$5,844	\$0	9,104	\$42,000	7	14%
5.3.2	Replace Shower Heads with Low Flow	0	0	18,881	3,468	\$20,189		36,271	\$34,000	2	59%
5.4.1	Replace Urinals Flush Valves	0	0	0	34.675	\$158	\$0	-	\$19,000	120	1%
5.4.2	Replace Water Closets with Ultra Low Flow	0	0	0	328.5	\$338	\$0	-	\$396,000	-	0%
5.4.3	Install Occupancy Sensors in the Stairwells T8 Fixtures	14,418	0	0	0	\$1,649	\$429	367	\$8,600	4	24%
5.4.4	Install Occupancy Sensors in the Fitness Room	8,854	0	0	0	\$1,013	\$0	225	\$1,100	1	92%
5.5.1	Replace Hot water boilers with Condensing Boiler at the end of useful service life	0	0	76,288	0	\$95,708	\$0	150,573	\$1,502,000	16	6%
5.5.2	Install HRV units on ACS 1, 4, 5, 6 AHUs	0	0	60,885	0	\$14,299	\$0	120,171	\$100,000 **	6	18%
5.5.3	Install HRV System with Kitchen Ecology Units KEU-1,2,3,4&5	0	0	72,418	0	\$17,008	\$0	142,933	\$95,000 **	5	20%
		23,272	-	233,212	4,871	\$ 156,207	429	459,645	2,197,000	14	7%

*Net investment includes opinions of probable cost (including tax) and estimated available incentives if applicable (incentives are one-time payments, contingent on third party approval; programs may change without notice).

**Incremental Costs



6. RECOMMENDATIONS

The energy audit shows that an annual energy savings of approximately 150,097 kWh of electricity, 154,838 m³ of natural gas per year, and 2,294 m³ of water per year. Energy efficiency measures are estimated to save approximately \$148,839 in utility costs and operating costs per year, and approximately 452,239 kg of CO₂ equivalent emissions per year.

We recommend that:

1. Energy efficiency measures 5.3.1, 5.3.2, 5.4.3 and 5.4.4 to be implemented at this time due to their relatively low capital costs and short payback periods.
2. Incorporate measures 5.5.2 and 5.5.3 into capital planning for implementation should they have an attractive payback based on Four Season Hotel investment criteria.
3. Incorporate measures 5.4.1 to 5.4.2 and 5.5.1 into long-term facility planning to be considered as end of life measures.

If you have any questions regarding the information contained herein, please contact the undersigned.

Sincerely,
Morrison Hershfield Limited



Marlon Katukurunda, E.I.T., CEM
Building Energy Consultant



Md Maniruzzaman Akan, M.A.Sc., P.Eng., CEM
Building Energy Engineer

APPENDIX A: Equipment List

Fan List												
Tag	Description	Service	Location	Nameplate		Drive	Estimated % Load ed	Estimated Input Power (kW)	Hours per Week	Hours per Year	Estimated Electricity Consumed (kWh)	Notes
				Flow (L/s)	Motor (HP)							
ACS-1	Supply	Basement BOH P2, P1	P2	7,551	24.9	VFD	50%	9.3	168	8760	81,501	52% Fresh Air. 24/7. Humidifier
ACS-4	Supply	Ground floor Hotel Lobby & Residential Lobby	2nd Floor	7,079	24.9	VFD	50%	9.3	168	8760	81,501	33% Fresh Air. 24/7. Humidifier
ACS-5	Supply	Ground floor Hotel Lobby & Residential Lobby	2nd Floor	3,304	15.0	VFD	50%	5.6	168	8760	49,076	28% Fresh Air.24/7. Humidifier
ACS-6	Supply	L3 Meeting Rooms	2nd Floor	9,438	30.0	VFD	50%	11.2	168	8760	98,151	40 % Fresh Air. 24/7. Humidifier
ACS-2	Toilet EF	TE-2	P2		2.0	Constant	85%	1.3	168	8760	11,173	100% Fresh Air. Humidifier
	Kitchen EF	SSE-1			0.5	Constant	85%	0.3	84	4380	1,378	
	General EF	ACE-2			7.5	VFD	85%	4.8	168	8760	41,640	
	Supply Fan	Basement BOH P2	P2	8,023	30.0	VFD	65%	14.6	168	8760	127,597	
ACS-3	Supply	L1, L2 Bar & Restaurant	P2	11,799	40.0	VFD	65%	19.4	84	4380	84,875	40% Fresh Air. Humidifier
	Recirc	ACRE-3			7.5	VFD	65%	3.6	84	4380	15,921	
ACS-7	Supply	L3 Ballroom & Pre-function	7th Floor	12,743	50.0	VFD	65%	24.3		2184	52,972	50% Fresh Air. HRV
	Recirc			12,743	50.0		65%	24.3		2184	52,972	
ACS-9	Supply			7,079	30.0	VFD	65%	14.6		2184	31,812	50% Fresh Air. HRV
	Recirc			7,079	15.0		85%	9.5		2184	20,800	
ACS-8	Supply	L4 BOH	7th Floor	2,360	10.1	VFD	65%	4.9		1560	7,608	30% Fresh Air
	Recirc						85%	0.0		0	0	
ACS-10	Supply	L6 Pre-function & Meeting Rooms	7Th Floor	4,719	15.0	VFD	65%	7.3		1040	7,574	50% Fresh Air
	Recirc						85%	0.0		0	0	
ACS-11	Supply	L8 Fitness	7Th Floor	4,719	15.0	VFD	65%	7.3		4380	31,899	33% Fresh Air
	Recirc						85%	0.0		0	0	

ACS-12	EF	L9 SPA	7Th Floor									
	EF						85%	0.0		0	0	
	EF						85%	0.0		0	0	
	Supply Fan			11,799	40.0	VFD	65%	19.4		4380	84,875	100% Fresh Air. HRV
	Return Fan			11,799	40.0		85%	25.3		4380	110,990	
ACS-13	EF						85%	0.0		0	0	
	Supply Fan	L9 Pool	7Th Floor	7,079	24.9	VFD	85%	15.8		8760	138,551	100% Fresh Air. HRV
	Return Fan			5,994	10.1		85%	6.4		8760	55,867	
ACS-14	Supply	L19 & L20 Royal Suite & Presidential Suite	20Th floor	1,227	5.0	Constant	85%	3.1		4380	13,781	33% Fresh Air
	Recirc			1,227	2.0	Constant	85%	1.3		4380	5,587	
ACS-15	Supply	L19 & L20 Royal Suite & Presidential Suite	20Th floor	1,227	5.0	Constant	85%	3.1		4380	13,781	33% Fresh Air
	Recirc			1,227	2.0	Constant	85%	1.3		4380	5,587	
ACS-16	Supply	2 Bedroom Suites	20Th floor	566	3.0	Constant	85%	1.9		4380	8,194	42% Fresh Air
	Recirc			566	1.0	Constant	85%	0.6		4380	2,793	
ACS-17	Supply	2 Bedroom Suites	20Th floor	566	3.0	Constant	85%	1.9		4380	8,194	42% Fresh Air
	Recirc			566	1.0	Constant	85%	0.6		4380	2,793	
MUA 1		Air Lock	P3	1,321	3.0	Constant	85%	1.9		1456	2,773	100% Fresh Air
MUA 2		Retail	Ground	706	3.0	Constant	85%	1.9		4368	8,171	
MUA 3	SF	L2 Main Kitchen	2nd Floor	2,926	10.1	VFD	65%	4.9		4368	21,303	
	EF	DE-2			0.5	Constant	80%	0.3		4368	1,293	
	EF	KEU-3			10.1	VFD	65%	4.9		4368	21,303	
MUA 4	SF	Ballroom Kitchen, Banquet Kitchen	5th Floor	2,643	7.5	VFD	65%	3.6		4368	15,906	

	EF	DE-3			0.5	Constant	80%	0.3		4368	1,293	
	EF	KEU-4			10.1	Constant	80%	6.0		4368	26,219	
MUA 5	SF	L6 Prep Kitchen	7th Floor	3,068	10.1	VFD	65%	4.9		4368	21,303	100% Fresh Air
	EF	KEU-5			15.0	Constant	80%	9.0		4368	39,153	
MUA 6	SF	Hotel Corridors	20th Floor	9,911	40.0	VFD	65%	19.4		8760	169,749	100% Fresh Air. HRV
	EF							0.0				
MUA 7	SF	Hotel Corridors	20th Floor	9,911	40.0	VFD	65%	19.4		8760	169,749	100% Fresh Air. HRV
	EF							0.0		8760	0	
KEU-1		From bakery			2.950		80%	1.8		4368	7,691	
KEU-2	EF	KEU-2			7.510		80%	4.5		4368	19,577	
	EF	DE-1			0.3		80%	0.2		4368	874	
LSF-1		P2 Laundry Supply Fan	Parking Level P2	5,191				5.59		1460	8,161	
BRS-1		Boiler Room Supply	Penthouse	708				0.56		1460	818	
RTS-2		Electrical Room Exhaust Fan	Mid-Level Mechanical room	7,551				3.37		1460	4,920	
MRS-1,3		P2 Mech RM Supply	P2, L7	8				0.37		1460	540	
MRS-4		L2 Mech RM Supply	L2 Mech RM	472				0.37		1460	540	
MRS-5		L5 Mech RM Supply	L5 Mech RM	236				0.25		1460	365	
MRS-6		Mid Lev Mech RM	MID Mech RM	949				0.56		1460	818	
MRS-8,9		Mid Lev North	Mid UI Fiortij	472				0.25		1460	365	
MRS-JO		Crease RM Ernd		661				0.38		1460	555	
MRS-11		Air Lock Press		142				0.25		1460	365	
MRS-12		Boiler Room Supply Fan	P3 Level	1,911				1.12		1460	1,635	
PGE-1		Parking level P1 Exhaust Fan	Parking Level P1	14,158				11.18		4380	48,968	

PGE-2		Parking level P1 Exhaust Fan	Parking Level P1	9,439				5.59		4380	24,484	
PGE-3		Parking level P1 Exhaust Fan	Parking Level P1	4,719				3.73		4380	16,337	
PGE-4		Parking level P2 Exhaust Fan	Parking Level P2	4,719				3.73		4380	16,337	
PGE-5		Parking level P2 Exhaust Fan	Parking Level P2	6,607				3.73		4380	16,337	
Total								376.0			1,917,348	

Pump List												
Tag	Description	Location	Quantity	Nameplate			Drive	Estimated % Loaded	Estimated Input Power (kW)	Hours per Year	Estimated Electricity Consumed (kWh)	Notes
				Flow (L/s)	Motor (KW)	Motor (HP)						
P-25	Condenser Pumps				56	74.96	Constant	80%	44.7	6602	295,360	On only in mechanical cooling. Assume serving lead chiller
P-26	Condenser Pumps				56	74.96	Constant	80%	44.7	3301	147,680	On only in mechanical cooling. Assume serving secondary chiller during peak cooling
P-19	Chilled Water Primary	P2 Chiller Room	1		30	39.96	Constant	80%	23.8	6602	157,455	Will be on when there's no free cooling
P-20	Chilled Water Primary	P2 Chiller Room	1		30	39.96	Constant	80%	23.8		0	Will be on when there's no free cooling
P-21	Chilled water supply pumps	P2 Chiller Room	1		37	50.02	VFD	50%	18.7	1445	26,968	P23/24 will operate lead lag. These pumps serve the ACS units. Assume it is on only for 8 months of the year
P-22	Chilled water supply pumps	P2 Chiller Room	1		37	50.02	VFD	50%	18.7	1445	26,968	P23/24 will operate lead lag. These pumps serve the ACS units. Assume it is on only for 8 months of the year
P-23	Chilled water supply pumps	P2 Chiller Room	1		19	24.94	VFD	50%	9.3	1445	13,448	P23/24 will operate lead lag. These pumps serve the fan coil units. Assume it is on only for 8 months of the year
P-24	Chilled water supply pumps	P2 Chiller Room	1		19	24.94	VFD	50%	9.3	1445	13,448	P23/24 will operate lead lag. These pumps serve the fan coil units. Assume it is on only for 8 months of the year
P-27A	Free cooling pumps	P2 Chiller Room	1		22	30.03	VFD	65%	14.6	8760	127,597	These pumps remain on during mechanical cooling and free cooling. Therefore, running the entire year.
P-27B	Free cooling pumps	P2 Chiller Room	1		22	30.03	VFD	65%	14.6	8760	127,597	These pumps remain on during mechanical cooling and free cooling. Therefore, running the entire year.
P-27C	Free cooling pumps	P2 Chiller Room	1		22	30.03	VFD	65%	14.6	2158	31,433	This pump only runs during free cooling. Hours based on 2019 hourly data where temp is below 0°C

P-29	Secondary chilled water supply pumps	P2 Chiller Room	1		15	19.98	VFD	65%	9.7	4380	42,437	P29/30 will operate lead lag. These pumps serve the refrigeration loop. On 24/7
P-30	Secondary chilled water supply pumps	P2 Chiller Room	1		15	19.98	VFD	65%	9.7	4380	42,437	P29/30 will operate lead lag. These pumps serve the refrigeration loop. On 24/7
SB1FP-1	Steam boiler pump									4368	0	
SB1FP-2	Steam boiler pump									4368	0	
P-1A	HX 1A/1B- hotel heating VAV pumps	L4 Mech Room	1		6	7.51	VFD	65%	3.6	2912	10,604	Assume it is on only for 4 months of the year
P-1B		L4 Mech Room	1		6	7.51	VFD	65%	3.6	2912	10,604	Assume it is on only for 4 months of the year
P-5A	HX 5A/5B- glycol heating ground to L8 AHU &UH	P2 Chiller Room	1		15	19.98	VFD	65%	9.7	2912	28,214	Assume it is on only for 4 months of the year. Lead/Lag
P-5B		P2 Chiller Room	1		15	19.98	VFD	65%	9.7	0	0	Assume it is on only for 4 months of the year. Lead/Lag
P-4A	HX 4A/4B- glycol heating P1 to P5 AHU &UH	P3 Boiler Room	1		15	19.98	VFD	65%	9.7	2912	28,214	Assume it is on only for 4 months of the year
P-4B	HX 4A/4B- glycol heating P1 to P5 AHU &UH	P3 Boiler Room	1		15	19.98	VFD	65%	9.7	2912	28,214	Assume it is on only for 4 months of the year
P-6A	HX 6A/6B- hotel heating									2912		Assume it is on only for 4 months of the year

	mid-level AHU											
P-6B	HX 6A/6B-hotel heating mid-level AHU									2912		Assume it is on only for 4 months of the year
P-7	HX 8/9-hotel heating FCU	Mid-Level Mech Room	1		11	15.019445	VFD	65%	7.3	2912	21,208	Assume it is on only for 4 months of the year. Lead/Lag
P-8	HX 8/9-hotel heating FCU	Mid-Level Mech Room	1		11	15.019445	VFD	65%	7.3	0	0	Assume it is on only for 4 months of the year. Lead/Lag
P9	Glycol-Hotel ACS & MUA	Mid-Level Mech Room	1		11	15.019445	VFD	65%	7.3	2912	21,208	Assume it is on only for 4 months of the year. Lead/Lag
P10	Glycol-Hotel ACS & MUA	Mid-Level Mech Room	1		11	15.019445	VFD	65%	7.3	0	0	Assume it is on only for 4 months of the year. Lead/Lag
P13	Hot water primary-DHW Boilers	Penthouse Boiler Room	1		11	15.019445	Constant	75%	8.4	4368	36,706	Lead/Lag. Assume running 12 hrs a day
P14/P15	Hot water primary-DHW Boilers	Penthouse Boiler Room	2		11	15.019445	Constant	75%	8.4	4368	36,706	Lead/Lag. Assume running 12 hrs a day
P16			1									
P17A	Hot water secondary-heating HEX	Penthouse Boiler Room	3		30	39.962451	VFD	75%	22.4	4368	97,664	
P17B	Hot water secondary-heating HEX	Penthouse Boiler Room	3		30	39.962451	VFD	75%	22.4	4368	97,664	
P17C	Hot water secondary-heating HEX	Penthouse Boiler Room	3		30	39.962451	VFD	65%	19.4	0	0	
Total									412.2		1,469,833	



Equipment List													
Tag	Description	Location	Quantity	Capacity	Nameplate		Drive	Estimated % Loaded	Estimated Input Power (kW)	Hours per Week	Hours per Year	Estimated Electricity Consumed (kWh)	Notes
					Flow (L/s)	Motor (HP)							
B-1	Heating boiler	4th Floor Mechanical Room	1	189.4					2928.0				85% Efficient
B-1	Heating boiler	4th Floor Mechanical Room	1						2928.0				85% Efficient
B-1	Heating boiler	4th Floor Mechanical Room	1						2928.0				85% Efficient
B-1	Heating boiler	4th Floor Mechanical Room	1						2928.0				85% Efficient
SB-1	Steam Boiler		1						644.2				81% Efficient
SB-2	Steam Boiler		1						480.0				96% Efficient
CP-1	Cooling Tower	Main Roof	1			3.00388 48							
CP-2	Cooling Tower	Main Roof	1			3.00388 48							
Total									12836			-	

APPENDIX B: Lighting List

Floor	Room/Area	Category/Schedule	Fixture Type	Quantity	Area (ft²)	NECB 2015 Value (watt/ft²)	Input Power (W)	Total Power (kW)	Annual Hours	Energy Consumption (kWh)
P2	Chiller Room	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	32			32	1	1,043	1,068
P2	Mechanical Room	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	21			32	1	1,043	701
P2	Women's Washroom	Typical 4 Hours lighting schedule	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	7			32	0	1,043	234
P2	Women's Lockers room	Typical 4 Hours lighting schedule	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	25			32	1	1,043	834
P2	Men's washroom	Typical 4 Hours lighting schedule	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	7			32	0	1,043	234
P2	Men's Lockers room	Typical 4 Hours lighting schedule	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	35			32	1	1,043	1,168
P2	House Keeping Room (Storage)	Typical- 6.00 AM- 6.00 PM	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	4			32	0	4,380	561
P2	Valet Laundry Room	24/7	Type F3 48" T8 Industrial Fluorescent 2LB	44			32	1	8,760	12,334
P2	House Keeping Room-02 (Storage)	Typical- 6.00 AM- 6.00 PM	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	4			32	0	4,380	561
P2	Laundry Room	24/7	Type F3 48" T8 Industrial Fluorescent 2LB	32			32	1	8,760	8,970
P2	Bulk Storage (Unused Textile)	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	8			32	0	4,380	1,121
P2	Storage -Stationery and flower work room	Typical 4 Hours lighting schedule	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	6			32	0	1,043	200

P2	Office Area	Office lighting	Type FA 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	10			32	0	5,397	1,727
P2		Office lighting	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	11			32	0	5,397	1,900
P2		Office lighting	Type A1 MR16 Downlight with Matt Al	14			8	0	5,397	567
P2	Telephone Room	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	4,380	140
P2	Electrical Room	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	4,380	140
P2	Stair	24/7	Type F2 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	2			32	0	8,760	561
P2	Elevator Lobby -P2-08	24/7	Type FA 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	2			32	0	8,760	561
P2	Storage - Night Cleaner's (P2-03)	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	4,380	140
P2	Compressor Room (P2-05)	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	2			32	0	4,380	280
P2	Corridor	24/7	Type FA 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	46			32	1	8,760	12,895
P2	Elevator Lobby -Service (P2- 04)	24/7	Type FA 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	2			32	0	8,760	561
P2	Elevator Pit	Typical 4 Hours lighting schedule	Type P Die casting Al Incandescent Light	10			100	1	1,043	1,043
P2	Elevator Lobby -P2-01	24/7	Type FA 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	3			32	0	8,760	841
P2	Dining Room - P2-32	Typical 4 Hours lighting schedule	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	28			32	1	1,043	934

P2		Typical 4 Hours lighting schedule	Type F4 2 X 4 T8 Clean Room Fluorescent Luminaire 2 LB	13			32	0	1,043	434
P2	Security Office -P2-36	24/7	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	7			32	0	8,760	1,962
P2		24/7	Type F3 48" T8 Industrial Fluorescent 2LB	3			32	0	8,760	841
P2	Storage - P2-38	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	6			32	0	1,043	200
P2		Typical 4 Hours lighting schedule	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	1			32	0	1,043	33
P2	Preparation and Storage -P2-33	Typical 4 Hours lighting schedule	Type F4 2 X 4 T8 Clean Room Fluorescent Luminaire 2 LB	46			32	1	1,043	1,535
P2	Elevator Lobby -P2-37	24/7	Type FA 2 X 4 T8 T Bar Ceiling Mtd Fluorescent Luminaire 2 LB	4			32	0	8,760	1,121
P2	Stair -P2-17	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	3			32	0	8,760	841
P2	Garbage Rooms-P2-41,42,43	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	39			32	1	4,380	5,466
P2	Maintenance Shop -P2-40	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	41			32	1	4,380	5,747
P2	Engineering Office -P2-40C,D	Office lighting	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	9			32	0	5,397	1,554
P2	Corridor -P2-41A	24/7	Type F3 48" T8 Industrial Fluorescent 2LB	5			32	0	8,760	1,402
P1	Electrical High Voltage Room -P1-30	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	24			32	1	1,043	801
P1	Elevator Lobby -P1-31,31A	24/7	Type FA 2 X 4 T8 T Bar Ceiling Mtd Fluorescent Luminaire 2 LB	7			32	0	8,760	1,962

P1	Electrical Room, Telecom - P1-28,29	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	16			32	1	1,043	534
P1	Office Rooms -P1-27	Office lighting	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	24			32	1	5,397	4,145
P1	Transformer Room -P1-34	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	33			32	1	1,043	1,101
P1	Corridor -P-02C,	24/7	Type FA 2 X 4 T8 T Bar Ceiling Mtd Fluorescent Luminaire 2 LB	24			32	1	8,760	6,728
P1	Vestibule - P1-17A	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	4			32	0	8,760	1,121
P1	Receiving Office -P1-35	Typical- 6.00 AM- 6.00 PM	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	3			32	0	4,380	420
P1	Washroom - P1-08	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
P1	Storage -P1-24A,B	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	3			32	0	1,043	100
P1	Computer Room -P1-23	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	17			32	1	4,380	2,383
P1	Telecom Room -P1-19,20	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	16			32	1	4,380	2,243
GF	Gas Room Utility Room) - GW-59	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	2			32	0	4,380	280
GF	Stair - GW-17	24/7	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	2			32	0	8,760	561
GF	Office - Receiving - GW-19	Typical- 6.00 AM- 6.00 PM	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	2			32	0	4,380	280
GF	Elevators -LS14,15,16,17	24/7	Not Available	0			0	0	8,760	0

GF	Elevator Lobby - GW-18	24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	3			32	0	8,760	841
GF	Stair - GW-16	24/7	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	3			32	0	8,760	841
GF	Loading Bay - GW-21	Typical- 6.00 AM- 6.00 PM	Type 01 Ceiling Hanged Garage Fixture	14			150	2	4,380	9,198
GF		Typical- 6.00 AM- 6.00 PM	Type 02 Ceiling Hanged Garage Fixture with Quartz	9			200	2	4,380	7,884
GF	Parking Ram - GW-22	24/7	Type FP T5 4 Feel Long Surface Mount Light 2LB	15			108	2	8,760	14,191
GF	Office Concierge -GW-42, 42A, 42B, 43	Office lighting	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	9			32	0	5,397	1,554
GF		Office lighting	Type F7 7 3/8" Lens Downlight 2LB	4			36	0	5,397	777
GF	Stair-GW-14	24/7	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	5			32	0	8,760	1,402
GF	Vest and Doorman Stand - GW-15A, 13	24/7	Type F7 7 3/8" Lens Downlight 2LB	2			36	0	8,760	631
GF		Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	4,380	140
GF	Stair -GW-07	24/7	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	2			32	0	8,760	561
GF	Office-Hotel Valet-GW-09,08	Office lighting	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	3			32	0	5,397	518
GF	Elevator Lobby -Luggage-GW-24	24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
GF	Stair -GW-02	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	4			32	0	8,760	1,121

GF	Corridor - GW-50, 04	24/7	Type FA 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	4			32	0	8,760	1,121
GF	Stair Corridor -GW-01A	24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	5			32	0	8,760	1,402
GF	Stair - GW- 01, 06	24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	3			32	0	8,760	841
GF	Pantry - GW-56	Typical- 6.00 AM- 6.00 PM	Type F4 2 X 4 T8 Clean Room Fluorescent Luminaire 2 LB	13			32	0	4,380	1,822
2	Stair M	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	3			32	0	8,760	841
2	Mechanical Room	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	26			32	1	1,043	868
2	Corridor	24/7	Type FA 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	4			32	0	8,760	1,121
2	Office - GM, Manager	Office lighting	Type A1 MR16 Downlight with Matt Al	12			8	0	5,397	486
2	Office -General Office -2L1-2	Office lighting	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	58			32	2	5,397	10,016
2		Office lighting	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	5,397	194
2	Board Room	Typical 4 Hours lighting schedule	Type A1 MR16 Downlight with Matt Al	8			8	0	1,043	63
2	Stair	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	4			32	0	8,760	1,121
2	IT Room	Typical- 6.00 AM- 6.00 PM	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	5			32	0	4,380	701

2	Corridor	24/7	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	1			32	0	8,760	280
2		24/7	Type F7 7 3/8" Lens Downlight 2LB	5			36	0	8,760	1,577
2	Storage	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
2	Washroom	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	4			36	0	1,043	150
2	Stair H, J	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	4			32	0	8,760	1,121
2	Stair L	24/7	Type F7 7 3/8" Lens Downlight 2LB	4			36	0	8,760	1,261
2		24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	1			32	0	8,760	280
2	Kitchen Area	Kitchen Lighting	Type F4 2 X 4 T8 Clean Room Fluorescent Luminaire 2 LB	55			32	2	5,475	9,636
2	Elevator Lobby - Service Elev.	24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	3			32	0	8,760	841
3	Stair - 3W-17	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	7			32	0.224	8760	1,962
3	Corridor	24/7	Type F7 7 3/8" Lens Downlight 2LB	11			36	0	8,760	3,469
3		Typical- 6.00 AM- 6.00 PM	Type A1 MR16 Downlight with Matt Al	1			8	0	4,380	33
3	Ballroom Kitchen	Kitchen Lighting	Type F4 2 X 4 T8 Clean Room Fluorescent Luminaire 2 LB	33			32	1	5,475	5,782
3	Stair L - 3W-16	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561

3	Stair H - 3W-14	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	4			32	0	8,760	1,121
3	Stair - 3W-01	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
3	Elevator Lobby - 3W-04	24/7	Type FA 2 X 4 T8 T Bar Ceiling Mtd Fluorescent Luminaire 2 LB	3			32	0	8,760	841
3	IT Room - 3W-08, 09	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	3			32	0	4,380	420
3	Pantry - 3W-11	Typical- 6.00 AM- 6.00 PM	Type F4 2 X 4 T8 Clean Room Fluorescent Luminaire 2 LB	6			32	0	4,380	841
3	Lobby Near Stair H 3W-14	24/7	Type F7 7 3/8" Lens Downlight 2LB	4			36	0	8,760	1,261
4	Stair - 4W-17	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	3			32	0.096	8760	841
4	Washroom - 4W-18	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	4			36	0	1,043	150
4	Elevator Lobby	24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
4	Stair - 4W-14	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
4	Stair - 4W-16	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
4	Mech Room - 4W-15B	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	12			32	0	1,043	400
4	Stair - 4W-01	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
4		24/7	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	8,760	315

4	Elevator Lobby-Service Elev.	24/7	Type FA 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	2			32	0	8,760	561
4		24/7	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	8,760	315
4	Storage - House Keeping 4W-09	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	3			36	0	1,043	113
4	Washroom - 4W-10	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
4	Elec Room - 4W-06	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	1,043	33
4	Ice Machine- 4W- 07	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
5	Stair - 5W-17	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
5	Storage 5W-28	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	1,043	33
5	Mechanical Room - 5W-26	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	19			32	1	1,043	634
5	Corridor and Elev Lobby - 5W-15	24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	13			32	0	8,760	3,644
5	Stair - 5W-16	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
5	Media / AV room - 5W-25	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	2			32	0	1,043	67
5	Storage - 5W-23, 24	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	10			32	0	1,043	334
5	Storage - 5W-21	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	3			32	0	1,043	100

5	Office Storage - 5W-19,20	Typical- 6.00 AM- 6.00 PM	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	6			32	0	4,380	841
5	Corridor 5W-15A	24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	6			32	0	8,760	1,682
5	IT Room - 5W-18A,C	Typical- 6.00 AM- 6.00 PM	Type F3 48" T8 Industrial Fluorescent 2LB	6			32	0	4,380	841
5	Fire Pump - 5W-18B	24/7	Type F3 48" T8 Industrial Fluorescent 2LB	9			32	0	8,760	2,523
5	Stair - 5W-14	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
5	Stair - 5W-01	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
5	Elevator Lobby - 5W-04	24/7	Type FA 2 X 4 T8 T Bar Ceiling Mtd Fluorescent Luminaire 2 LB	2			32	0	8,760	561
5	Storage - House Keeping 5W-05	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
5	Storage - House Keeping, Jani 5W-09	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	3			36	0	1,043	113
5	Elec and Tel Room - 5W-06,10	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	2			32	0	1,043	67
5	Ice Machine- 5W-07	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
6	Stair M- 6W-17	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
6	Washroom-Staff 6W-25	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
6	Corridor 6W-26	24/7	Type F 2 X 4 T8 Fluorescent Luminaire 2 LB	3			32	0	8,760	841

6		24/7	Type F7 7 3/8" Lens Downlight 2LB	7			36	0	8,760	2,208
6		24/7	Type A2 Wall Mounted CFL	3			13	0	8,760	342
6	Kitchen- Prep- 6W-24	Kitchen Lighting	Type F4 2 X 4 T8 Clean Room Fluorescent Luminaire 2 LB	25			32	1	5,475	4,380
6	Stair L - 6W-16	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
6	Stair H - 6W-14	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
6	Stair A - 6W-01	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
6	Elevator Lobby- Service Elev. 6W-04	24/7	Type FA 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	2			32	0	8,760	561
6	Storage- House Keeping 6W- 05	24/7	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	8,760	315
6	Storage- House Keeping 6W- 09	24/7	Type F7 7 3/8" Lens Downlight 2LB	3			36	0	8,760	946
6	Washroom - 6W-10	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
6	Elec Room - 6W-06	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	1,043	33
6	Ice Machine- 6W-07	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
7	Void	Typical 4 Hours lighting schedule	Type P Die casting Al Incandescent Light	2			100	0	1,043	209
7	Corridor - 7W-15	24/7	Type F3 48" T8 Industrial Fluorescent 2LB	14			32	0	8,760	3,924

7		24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	18			32	1	8,760	5,046
7	Elevator Lobby - 7W-15C	24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	5			32	0	8,760	1,402
7	Mechanical Room - 7W-19	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	33			32	1	1,043	1,101
7	Stair H - 7W-14	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
7	Stair L - 7W-16	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
7	Storage - 7W-17, 22	Typical 4 Hours lighting schedule	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	6			32	0	1,043	200
7	IT and Elec Room - 7W-20,21	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	6			32	0	1,043	200
7	Storage - BOH 7W-18	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	16			32	1	1,043	534
7	Stair B - 7W-02	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
7	Elevator Lobby- Service Elev. 7W-04	24/7	Type FA 2 X 4 T8 T Bar Ceiling Mtd Fluorescent Luminaire 2 LB	2			32	0	8,760	561
7	Storage- House Keeping 7W-05	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
7	Storage- House Keeping 7W-08, 09	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	3			36	0	1,043	113
7	Tel Room - 7W-10	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	1,043	33
7	Elec Room - 7W-06	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	1,043	33

7	Ice Machine- 7W-07	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
8	Elevator Machine Room - 8W-26	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	2			32	0	1,043	67
8	Stair H - 8W-14	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
8	Stair L - 8W-16	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
8	Stair L - 8W-14	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	1			32	0	8,760	280
8	Stair L - 8W-01	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
8	Elevator Machine Room - 8W-25	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	6			32	0	1,043	200
8	Mechanical Room -8W-24	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	8			32	0	1,043	267
8	Storage BOH-8W-17	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	10			32	0	1,043	334
8	Elevator Lobby- Service Elev. 8W-04	Typical 4 Hours lighting schedule	Type FA 2 X 4 T8 T Bar Ceiling Mtd Fluorescent Luminaire 2 LB	2			32	0	1,043	67
8	Storage- House Keeping 8W-05	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
8	Storage- House Keeping 8W-08, 09	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	3			36	0	1,043	113
8	Washroom - 8W-10	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
8	Elec Room - 8W-06	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	1,043	33

8	Ice Machine- 8W-07	Typical 4 Hours lighting scheduled	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
8	Corridor -Stair H 8W-14A	24/7	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
9	Stair L - 9W-16	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
9	Stair H - 9W-14	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
9	Stair L - 9W-2	24/7	Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB	2			32	0	8,760	561
9	Elevator Lobby- Service Elev. 9W-04	24/7	Type FA 2 X 4 T8 T Bar Celling Mtd Fluorescent Luminaire 2 LB	2			32	0	8,760	561
9	Storage- House Keeping 9W-05	Typical 4 Hours lighting schedule	Type F7 7 3/8" Lens Downlight 2LB	1			36	0	1,043	38
9	Elec Room - 9W-06	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	1			32	0	1,043	33
9	Storage BOH- 9W-14	Typical 4 Hours lighting schedule	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	6			32	0	1,043	200
9	Steam Room -9W-11U, 11V	Typical 4 Hours lighting schedule	Type F1 48" T8 Wrap Surface Mounted Fluorescent 2LB	4			32	0	1,043	133
MMR	Mechanical Room	Typical 4 Hours lighting schedule	Type F3 48" T8 Industrial Fluorescent 2LB	134			32	4	1,043	4,472
P	Parking P1 for 55 Parking spot and 103 FP fixtures (50 Watt)	Typical- 6.00 AM- 6.00 PM	Type FP T5 4 Feet Long Surface Mount Light 2LB		33091	0.26		9	4,380	37,322
9	Pool -9W-17 NECB 2015	Typical- 6.00 AM- 6.00 PM			4347	0.72		3	4,380	13,798
9	Pedicure -9W-19 NECB 2016	Typical- 6.00 AM- 6.00 PM			1549	0.91		1	4,380	6,177

9	Change room - Women's - 9W-12 NECB 2015	Typical 4 Hours lighting schedule			3337	0.98		3	1,043	3,395
9	Treatment Rooms -9W-11- NECB 2015	Typical- 6.00 AM- 6.00 PM			5414	0.90		5	4,380	21,368
9	Corridor - 9W-08,09 -NECB 2015	24/7			1614	0.27		0	8,760	3,818
9	Change room - Men's -9W-13 - NECB 2015	24/7			3312	0.98		3	8,760	28,303
9	Public Spa -NECB 2015	24/7			586	0.90		1	8,760	4,628
9	Reception Retail - 9W-10- NECB 2015	Typical- 6.00 AM- 6.00 PM			428	0.87		0	4,380	1,639
9	Corridor - 9W-16A -NECB 2015	24/7			1465	0.27		0	8,760	3,465
9	Stair -9W-18 - NECB 2015	24/7			203	0.69		0	8,760	1,225
9	Washroom -9W-16B, C NECB 2015	Typical 4 Hours lighting schedule			117	0.98		0	1,043	120
10	Elevator Lobby- Service Elev. 10W-04 NECB 2015	24/7			2497	1.06		3	8,760	23,371
10	Storage- House Keeping 10W-05 NECB 2015	Typical 4 Hours lighting schedule			676	0.63		0	1,043	445
10	Storage- House Keeping 10W-08, 09	Typical 4 Hours lighting schedule			1124	0.63		1	1,043	740
10	Tel Room - 10W-10 NECB 2015	Typical 4 Hours lighting schedule			714	0.42		0	1,043	318
10	Elec Room - 10W-06 NECB 2015	Typical 4 Hours lighting schedule			704	0.43		0	1,043	314
10	Ice Machine- 10W-07 NECB 2015	Typical 4 Hours lighting schedule			459	0.43		0	1,043	205

10	Corridor NECB 2015	24/7			1527 7	0.27		4	8,760	36,127
GF	Lounge and Bar NECB 2015	Typical- 6.00 AM- 6.00 PM			6,072	0.74		4	4,380	19,521
GF	Reception Area NECB 2015	24/7			2294	0.87		2	8,760	17,552
GF	Guest Sitting Area NECB 2015	Typical- 6.00 AM- 6.00 PM			1369	0.74		1	4,380	4,402
GF	Corridor - GW-27, 23 NECB 2015	24/7			697	0.27		0	8,760	1,648
GF	Artwork and Shop GW-25, 26 NECB 2015	Typical 4 Hours lighting schedule			319	0.61		0	1,043	204
2	Office Area - Business Center, Admin and Acc. NECB 2015	Office lighting			775	0.98		1	5,397	4,118
2	Washrooms NECB 2015	24/7			929	0.98		1	8,760	7,944
2	Lounge and seating area NECB 2015	Typical- 6.00 AM- 6.00 PM			3780	0.73		3	4,380	12,154
2	Seating Area NECB 2015	Typical- 6.00 AM- 6.00 PM			4059	0.73		3	4,380	13,049
2	Elevator Lobby NECB 2015	24/7			293	0.27		0	8,760	693
3	Ballroom and Preparation rooms NECB 2015	Typical 4 Hours lighting schedule			1035 6	1.06		11	1,043	11,438
3	Lobby Corridor 3W-03 NECB 2015	24/7			5138	0.27		1	8,760	12,150
3	Women's Washroom 3W-05 NECB 2015	24/7			866	0.98		1	8,760	7,404
3	Men's Washroom 3W-13 NECB 2015	24/7			884	0.98		1	8,760	7,557

3	Meeting and Board Room 3W-18, 19 NECB 2015	9.00 AM -10.00 AM			2041	1.26		3	261	658
3	Meeting and Board Room 3W-06,07 NECB 2015	9.00 AM -10.00 AM			2041	1.24		3	261	658
3	Corridor 3W-10, 12 NECB 2015	24/7			1089	0.27		0	8,760	2,578
3	Closet – Coats NECB 2015	24/7			239	0.63		0	8,760	1,320
3	Meeting Room 3W-22 NECB 2015	9.00 AM -10.00 AM			270	1.24		0	261	87
4	Corridor and Lobby 4th Floor NECB 2015	24/7			1552	0.27		0	8,760	3,671
5	Corridor and Lobby 5th Floor NECB 2015	24/7			1552	0.27		0	8,760	3,671
6	Ball room 6W-28 NECB 2015	Typical 4 Hours lighting schedule			6869	1.06		7	1,043	7,586
6	Meeting Room 6W-29 NECB 2015	9.00 AM -10.00 AM			1465	1.24		2	261	472
6	Corridor 6W-26 NECB 2015	24/7			413	0.27		0	8,760	978
6	Corridor 6W-03, 11, 12,13 NECB 2015	24/7			1525	0.27		0	8,760	3,607
6	Washroom -6W-18,19 NECB 2015	24/7			984	0.98		1	8,760	8,413
7	Corridor 7W-11,12,13 NECB 2015	24/7			1540	0.27		0	8,760	3,642
8	Fitness Room NECB 2015	24/7			2789	0.72		2	8,760	17,708
8	Corridor 8W-15C NECB 2015	24/7			1597	0.27		0	8,760	3,776

8	Corridor 8W-11,12,13 NECB 2015	24/7			1556	0.27		0	8,760	3,681
8	Washroom - Barrier free 8W-19, 20 NECB 2015	Typical 4 Hours lighting schedule			156	0.98		0	1,043	159
Suites	Standard room NECB 2015	Typical Suites Occu. lighting 12.00PM - 24:00PM			82965	0.65		54	4,380	238,005
Suites	Standard barrier free NECB 2015	Typical Suites Occu. lighting 12.00PM - 24:00PM			15482	0.65		10	4,380	44,415
Suites	Standard Double Bed NECB 2015	Typical Suites Occu. lighting 12.00PM - 24:00PM			16827	0.65		11	4,380	48,273
Suites	Executive NECB 2015	Typical Suites Occu. lighting 12.00PM - 24:00PM			27648	0.65		18	4,380	79,315
Suites	1 Bedroom NECB 2015	Typical Suites Occu. lighting 12.00PM - 24:00PM			3308	0.65		2	4,380	9,491
Suites	2 Bedroom NECB 2015	Typical Suites Occu. lighting 12.00PM - 24:00PM			1030	0.65		1	4,380	2,955
Suites	Royal Suite NECB 2015	Typical Suites Occu lighting 12.00PM - 24:00PM			2635	0.65		2	4,380	7,559
Suites	Presidential Suite NECB 2015	Typical Suites Occu lighting 12.00PM - 24:00PM			2120	0.65		1	4,380	6,082

APPENDIX C: Utility Bill Records



Month	Electricity									Natural Gas			Water & Sewage		Totals					Weather Data	Weather Data
	Adjusted Consumption (GJ)	Adjusted Consumption (kWh)	Adjusted Cost (\$)	Demand kW	Demand kVa	Average Demand (kW)	Load Factor %	Power Factor %	GHG Emissions (tGHG)	Adjusted Consumption (GJ)	Adjusted Cost (\$)	GHG Emissions (tGHG)	Adjusted Consumption (m³)	Adjusted Cost (\$)	Total Adj. Consumption (GJ)	Total Adj. Cost (\$)	Total GHG Emissions (tCO2)	Energy Intensity (GJ/m²)	Cost Index (\$/m²)	HDD base 10.°C	CDD base 13.°C
Jan-17	2,956	821172	\$101,805	0	0	1103.7264	#DIV/0!	#DIV/0!	41.1	5,146	\$ 32,503	255.8	6,140	\$24,063	8,103	\$158,371	297	0.1	2.6544	357	0
Feb-17	2,689	747038	\$121,340	0	0	1111.6631	#DIV/0!	#DIV/0!	37.4	4,527	\$ 30,976	225.0	6,123	\$23,997	7,217	\$176,313	262	0.1	\$2.96	289	1
Mar-17	2,974	826013	\$122,200	0	0	1110.2327	#DIV/0!	#DIV/0!	41.3	5,218	\$ 28,837	259.3	6,861	\$26,889	8,192	\$177,926	301	0.1	\$2.98	328	0
Apr-17	2,900	805573	\$93,101	0	0	1118.8512	#DIV/0!	#DIV/0!	40.3	3,156	\$ 14,117	156.8	6,863	\$26,897	6,056	\$134,115	197	0.1	\$2.25	74	27
May-17	3,111	864222	\$124,913	0	0	1161.5885	#DIV/0!	#DIV/0!	43.2	3,070	\$ 16,968	152.6	10,572	\$41,433	6,182	\$183,314	196	0.1	\$3.07	30	60
Jun-17	3,294	915056	\$147,698	0	0	1270.9105	#DIV/0!	#DIV/0!	45.8	2,610	\$ 15,795	129.7	9,301	\$36,452	5,904	\$199,945	175	0.1	\$3.35	0	195
Jul-17	3,564	990120	\$137,267	0	0	1330.8061	#DIV/0!	#DIV/0!	49.5	2,422	\$ 17,207	120.4	9,365	\$36,702	5,986	\$191,176	170	0.1	\$3.20	0	268
Aug-17	3,561	989192	\$134,487	0	0	1329.5592	#DIV/0!	#DIV/0!	49.5	2,587	\$ 17,697	128.5	9,378	\$36,753	6,148	\$188,938	178	0.1	\$3.17	0	225
Sep-17	3,474	965000	\$121,830	0	0	1340.2773	#DIV/0!	#DIV/0!	48.2	2,631	\$ 16,614	130.7	9,006	\$35,295	6,105	\$173,740	179	0.1	\$2.91	3	185
Oct-17	3,198	888330	\$126,519	0	0	1193.9918	#DIV/0!	#DIV/0!	44.4	2,838	\$ 20,164	141.0	8,133	\$31,874	6,036	\$178,557	185	0.1	\$2.99	31	78
Nov-17	2,905	806891	\$113,306	0	0	1120.6819	#DIV/0!	#DIV/0!	40.3	3,872	\$ 25,474	192.4	9,065	\$35,527	6,777	\$174,307	233	0.1	\$2.92	188	3
Dec-17	3,052	847725	\$130,723	0	0	1139.4151	#DIV/0!	#DIV/0!	42.4	5,819	\$ 45,942	289.2	7,004	\$27,449	8,871	\$204,114	332	0.1	\$3.42	458	0
Jan-18	3,140	872200	\$114,730	0	0	1172.3124	#DIV/0!	#DIV/0!	43.6	6,113	\$ 35,389	303.8	6,094	\$25,794	9,253	\$175,912	347	0.2	\$2.95	476	0
Feb-18	2,701	750233	\$100,389	0	0	1116.4178	#DIV/0!	#DIV/0!	37.5	4,757	\$ 28,791	236.4	6,511	\$27,559	7,458	\$156,739	274	0.1	\$2.63	329	1
Mar-18	2,911	808525	\$114,075	0	0	1086.7267	#DIV/0!	#DIV/0!	40.4	4,799	\$ 30,306	238.5	6,379	\$27,000	7,709	\$171,381	279	0.1	\$2.87	314	0
Apr-18	2,866	796168	\$116,369	0	0	1105.7882	#DIV/0!	#DIV/0!	39.8	4,214	\$ 23,289	209.4	6,379	\$27,000	7,080	\$166,658	249	0.1	\$2.79	211	5
May-18	3,297	915799	\$111,966	0	0	1230.913	#DIV/0!	#DIV/0!	45.8	2,823	\$ 16,346	140.3	8,171	\$34,585	6,120	\$162,897	186	0.1	\$2.73	5	150
Jun-18	3,366	934879	\$130,978	0	0	1298.4433	#DIV/0!	#DIV/0!	46.7	2,618	\$ 13,779	130.1	8,456	\$35,791	5,983	\$180,548	177	0.1	\$3.03	0	205
Jul-18	3,889	1080168	\$124,588	0	0	1451.8388	#DIV/0!	#DIV/0!	54.0	2,519	\$ 15,910	125.2	8,934	\$37,814	6,408	\$178,312	179	0.1	\$2.99	0	328
Aug-18	4,005	1112575	\$123,136	0	0	1495.3968	#DIV/0!	#DIV/0!	55.6	2,635	\$ 14,560	130.9	9,188	\$38,889	6,640	\$176,586	187	0.1	\$2.96	0	314
Sep-18	3,637	1010323	\$116,165	0	0	1403.2261	#DIV/0!	#DIV/0!	50.5	2,580	\$ 14,934	128.2	8,629	\$36,523	6,217	\$167,623	179	0.1	\$2.81	1	188
Oct-18	3,168	879993	\$118,980	0	0	1182.7857	#DIV/0!	#DIV/0!	44.0	3,390	\$ 21,408	168.5	7,665	\$32,443	6,558	\$172,831	212	0.1	\$2.90	84	29
Nov-18	2,885	801466	\$109,970	0	0	1113.1478	#DIV/0!	#DIV/0!	40.1	4,364	\$ 25,262	216.9	6,467	\$27,372	7,249	\$162,605	257	0.1	\$2.73	250	0
Dec-18	3,027	840830	\$108,019	0	0	1130.1473	#DIV/0!	#DIV/0!	42.0	4,702	\$ 29,697	233.7	6,644	\$28,122	7,729	\$165,838	276	0.1	\$2.78	308	0
Jan-19	3,086	857281	\$120,772	0	0	1152.26	#DIV/0!	#DIV/0!	42.9	6,173	\$ 38,986	306.8	5,879	\$26,874	9,259	\$186,633	350	0.2	\$3.13	511	0
Feb-19	2,780	772255	\$112,265	0	0	1149.1883	#DIV/0!	#DIV/0!	38.6	5,322	\$ 35,011	264.5	5,674	\$25,937	8,102	\$173,214	303	0.1	\$2.90	394	0
Mar-19	2,780	772255	\$115,838	0	0	1037.9765	#DIV/0!	#DIV/0!	38.6	4,916	\$ 25,874	244.3	6,032	\$27,574	7,696	\$169,286	283	0.1	\$2.84	340	0
Apr-19	2,882	800449	\$83,783	0	0	1111.735	#DIV/0!	#DIV/0!	40.0	3,425	\$ 27,042	170.2	6,148	\$28,104	6,307	\$138,929	210	0.1	\$2.33	128	5
May-19	3,129	869114	\$79,395	0	0	1168.1644	#DIV/0!	#DIV/0!	43.5	2,910	\$ 21,444	144.6	6,638	\$30,344	6,039	\$131,183	188	0.1	\$2.20	30	43
Jun-19	3,300	916692	\$85,909	0	0	1273.1835	#DIV/0!	#DIV/0!	45.8	2,549	\$ 11,406	126.7	7,678	\$35,098	5,850	\$132,412	173	0.1	\$2.22	1	170
Jul-19	3,909	1085763	\$96,925	0	0	1459.3594	#DIV/0!	#DIV/0!	54.3	2,387	\$ 14,450	118.7	8,895	\$40,661	6,296	\$152,037	173	0.1	\$2.55	0	326
Aug-19	3,622	1006200	\$105,152	0	0	1352.4195	#DIV/0!	#DIV/0!	50.3	2,619	\$ 17,923	130.2	7,567	\$34,591	6,242	\$157,665	180	0.1	\$2.64	0	259
Sep-19	3,312	919982	\$85,221	0	0	1277.7524	#DIV/0!	#DIV/0!	46.0	2,644	\$ 17,392	131.4	7,567	\$34,591	5,955	\$137,203	177	0.1	\$2.30	0	149
Oct-19	3,137	871466	\$89,130	0	0	1171.3256	#DIV/0!	#DIV/0!	43.6	2,927	\$ 20,027	145.5	6,712	\$30,682	6,064	\$139,839	189	0.1	\$2.34	35	26
Nov-19	2,899	805325	\$115,171	0	0	1118.5074	#DIV/0!	#DIV/0!	40.3	4,462	\$ 24,659	221.8	6,466	\$29,558	7,361	\$169,388	262	0.1	\$2.84	269	0
Dec-19	2,958	821688	\$119,525	0	0	1104.4193	#DIV/0!	#DIV/0!	41.1	4,926	\$ 16,852	244.8	6,697	\$30,614	7,884	\$166,990	286	0.1	\$2.80	331	0



Jan-20	2,892	803213	\$122,847	0	0	1079.5873	#DIV/0!	#DIV/0!	40.2	5,049	\$ 27,900	250.9	7,036	\$34,736	7,940	\$185,484	291	0.1	\$3.11	348	0
Feb-20	2,741	761484	\$122,100	0	0	1094.0858	#DIV/0!	#DIV/0!	38.1	4,992	\$ 28,900	248.1	6,365	\$31,424	7,733	\$182,424	286	0.1	\$3.06	368	0
Mar-20	2,304	640136	\$126,695	0	0	860.39718	#DIV/0!	#DIV/0!	32.0	3,261	\$ 12,016	162.1	3,551	\$17,531	5,566	\$156,242	194	0.1	\$2.62	218	2
Apr-20	1,639	455146	\$98,244	0	0	632.14662	#DIV/0!	#DIV/0!	22.8	1,610	\$ 8,473	80.0	732	\$3,614	3,248	\$110,331	103	0.1	\$1.85	132	1
May-20	1,838	510694	\$100,883	0	0	686.41702	#DIV/0!	#DIV/0!	25.5	1,387	\$ 6,570	68.9	1,743	\$8,605	3,226	\$116,058	94	0.1	\$1.95	57	82
Jun-20	2,118	588368	\$115,551	0	0	817.17803	#DIV/0!	#DIV/0!	29.4	980	\$ 4,643	48.7	2,215	\$9,497	3,098	\$129,692	78	0.1	\$2.17	1	238
Jul-20	2,839	788702	\$139,946	0	0	1060.0839	#DIV/0!	#DIV/0!	39.4	1,266	\$ 6,661	62.9	3,608	\$15,470	4,105	\$162,077	102	0.1	\$2.72	0	371
Aug-20	2,955	820898	\$131,416	0	0	1103.3575	#DIV/0!	#DIV/0!	41.0	1,616	\$ 9,354	80.3	4,289	\$18,390	4,571	\$159,161	121	0.1	\$2.67	0	286
Sep-20	2,763	767395	\$117,576	0	0	1065.8264	#DIV/0!	#DIV/0!	38.4	1,763	\$ 13,919	87.6	4,104	\$17,597	4,526	\$149,093	126	0.1	\$2.50	5	132
Oct-20	2,505	695962	\$118,030	0	0	935.4328	#DIV/0!	#DIV/0!	34.8	2,108	\$ 14,423	104.8	2,547	\$10,921	4,613	\$143,374	140	0.1	\$2.40	65	20
Nov-20	2,259	627519	\$111,052	0	0	871.55417	#DIV/0!	#DIV/0!	31.4	2,204	\$ 16,823	109.6	1,849	\$7,928	4,463	\$135,803	141	0.1	\$2.28	131	20
Dec-20	2,291	636469	\$119,161	0	0	855.46909	#DIV/0!	#DIV/0!	31.8	3,159	\$ 20,781	157.0	2,550	\$10,934	5,450	\$150,875	189	0.1	\$2.53	318	0
Jan-21	2,330	647315	\$106,991	0	0	870.04704	#DIV/0!	#DIV/0!	32.4	3,422	\$ 22,516	170.1	2,232	\$9,714	5,753	\$139,222	202	0.1	\$2.33	383	0
Feb-21	2,193	609245	\$83,256	0	0	906.61458	#DIV/0!	#DIV/0!	30.5	3,728	\$ 24,525	185.3	1,901	\$8,274	5,921	\$116,054	216	0.1	\$1.95	423	0
Mar-21	2,379	660906	\$108,413	0	0	888.31452	#DIV/0!	#DIV/0!	33.0	2,872	\$ 18,897	142.8	2,121	\$9,231	5,252	\$136,541	176	0.1	\$2.29	227	10
Apr-21	2,209	613720	\$103,599	0	0	852.38927	#DIV/0!	#DIV/0!	30.7	2,148	\$ 15,266	106.8	1,532	\$6,668	4,358	\$125,532	137	0.1	\$2.10	102	14
May-21	2,433	675956	\$105,810	0	0	908.54295	#DIV/0!	#DIV/0!	33.8	1,746	\$ 11,947	86.8	2,460	\$10,706	4,180	\$128,463	121	0.1	\$2.15	31	105
Jun-21	2,882	800566	\$121,753	0	0	1111.8974	#DIV/0!	#DIV/0!	40.0	1,479	\$ 10,896	73.5	3,252	\$14,153	4,361	\$146,803	114	0.1	\$2.46	0	267
Jul-21	3,268	907776	\$118,834	0	0	1220.1295	#DIV/0!	#DIV/0!	45.4	1,948	\$ 13,843	96.8	5,692	\$24,773	5,216	\$157,450	142	0.1	\$2.64	0	256
Aug-21	3,749	1041263	\$121,150	0	0	1399.547	#DIV/0!	#DIV/0!	52.1	2,201	\$ 15,637	109.4	7,715	\$33,577	5,949	\$170,364	161	0.1	\$2.86	0	328
Sep-21	3,115	865359	\$107,514	0	0	1201.8881	#DIV/0!	#DIV/0!	43.3	2,012	\$ 14,822	100.0	6,079	\$26,457	5,127	\$148,793	143	0.1	\$2.49	0	146
Oct-21	3,085	856840	\$98,286	0	0	1151.6665	#DIV/0!	#DIV/0!	42.8	2,531	\$ 20,648	125.8	5,930	\$25,809	5,616	\$144,743	169	0.1	\$2.43	23	78
Nov-21	2,653	736868	\$95,982	0	0	1023.4283	#DIV/0!	#DIV/0!	36.8	3,403	\$ 26,863	169.1	4,800	\$20,891	6,055	\$143,736	206	0.1	\$2.41	184	3
Dec-21	2,624	728823	\$101,023	0	0	979.6011	#DIV/0!	#DIV/0!	36.4	4,001	\$ 32,641	198.9	4,956	\$21,570	6,625	\$155,233	235	0.1	\$2.60	267	2
Jan-22	2,500	694490	\$89,281	0	0	933.45475	#DIV/0!	#DIV/0!	34.7	4,960	\$ 43,076	246.5	2,696	\$11,728	7,460	\$144,085	281				
Feb-22	2,431	675168	\$88,657	0	0	1004.7142	#DIV/0!	#DIV/0!	33.8	4,604	\$ 38,769	228.8	4,196	\$18,253	7,034	\$145,679	263				
Mar-22	2,762	767320	\$99,283	0	0	1031.3436	#DIV/0!	#DIV/0!	38.4	4,330	\$ 37,604	215.2	5,662	\$24,630	7,093	\$161,517	254				
Apr-22	2,730	758458	\$100,825	0	0	1053.4135	#DIV/0!	#DIV/0!	37.9	3,285	\$ 31,986	163.3	6,062	\$27,174	6,015	\$159,985	201				
May-22	3,117	865698	\$112,307	0	0	1163.573	#DIV/0!	#DIV/0!	43.3	2,471	\$ 24,706	122.8	7,112	\$31,881	5,587	\$168,894	166				
Jun-22	3,181	883628	\$120,660	0	0	1227.2616	#DIV/0!	#DIV/0!	44.2	2,201	\$ 22,010	109.4	8,298	\$37,197	5,382	\$179,868	154				
Jul-22	3,545	984690	\$116,586	0	0	1323.5084	#DIV/0!	#DIV/0!	49.2	2,153	\$ 27,200	107.0	8,877	\$39,793	5,698	\$183,579	156				
Aug-22	3,648	1013405	\$117,080	0	0	1362.1038	#DIV/0!	#DIV/0!	50.7	2,196	\$ 27,735	109.1	8,662	\$38,829	5,844	\$183,645	160				
Sep-22	3,217	893512	\$100,044	0	0	1240.9895	#DIV/0!	#DIV/0!	44.7	2,293	\$ 29,570	114.0	7,874	\$0	5,510	\$129,614	159				
Oct-22	0	0	\$0	0	0	0	#DIV/0!	#DIV/0!	0.0	2,869	\$ 38,501	142.6	7,370	\$0	2,869	\$38,501	143				
Nov-22	0	0	\$0	0	0	0	#DIV/0!	#DIV/0!	0.0	0	\$ -	0.0	0	\$0	0	\$0	0				
Dec-22	0	0	\$0	0	0	0	#DIV/0!	#DIV/0!	0.0	0	\$ -	0.0	0	\$0	0	\$0	0				
	Adjusted	Adjusted	Adjusted						Elec GHG	Adjusted	Adjusted	NG GHG	Adjusted	Adjusted	Total Adj.	Total Adj.	Total GHG	Energy	Cost	HDD	CDD
	Consumption	Consumption	Cost						Intensity	Consumption	Cost	Intensity	Consumption	Cost	Consumption	Cost	Intensity	Intensity	Index	base	base
	(GJ)	(kWh)	(\$)						(kgCO2/m2)	(GJ)	(\$)	(kgCO2/m2)	(m³)	(\$)	(GJ)	(\$)	(kgCO2/m2)	(GJ/m²)	(\$/m²)	10.°C	13.°C
2017	37,679	10,466,331	1,475,190						9	43,895	282,295	37	97,811	383,331	81,574	\$2,140,816	45	1.4		1,758	1,042



2018	38,891	10,803,158	1,389,364						9	45,512	269,671	38	89,517	378,893	84,403	\$2,037,928	47	1.4		1,976	1,220
2019	37,794	10,498,471	1,209,088						9	45,261	271,064	38	81,953	374,627	83,055	\$1,854,779	47	1.4		2,040	977
2020	29,146	8,095,986	1,423,502						7	29,394	170,462	24	40,589	186,648	58,540	\$1,780,613	31	1.0		1,644	1,151
2021	32,921	9,144,638	1,272,611	0	0		#DIV/0!		8	31,491	228,502	26	48,670	211,822	64,412	\$1,712,934	34	1.1		1,641	1,210
Avg 3yr	33,287	9,246,365	1,301,734							35,382	223,343		57,071	264,255	68,669	\$1,782,775					
12 mo	32,921	9,144,638	1,272,611						8	31,491	228,502	26	48,670	211,822	64,412	1,712,934	34	1		1,641	1,210



APPENDIX D: Weather Normalization

WEATHER NORMALIZATION

Weather normalization of energy consumption provides a method for verifying energy savings associated with energy efficiency initiatives — such as those recommended in this report.

After energy efficiency measures are implemented, their effectiveness may not be readily discerned by a direct comparison with past energy use due to the variability that is introduced by weather. The baseline year's energy consumption is correlated to degree days, below. Using these correlations, energy consumption of the baseline can be determined for other periods using the actual degree days from that period. This represents what the building performance would be without implementing energy efficiency measures and provides a way to verify that efficiency measures perform as expected.

Electricity

The weather sensitivity of the buildings' electricity consumption was examined by plotting the average outdoor air temperature against daily consumption for each month, shown in Figure 11.

There appears to be two trends evident. One during warm months above about 13°C, which would be dominated by mechanical cooling loads (e.g., chillers, compressors, fans). Below 13°C the electricity use shows a weaker relationship to outdoor temperatures due to less fan energy consumed during cooler months.

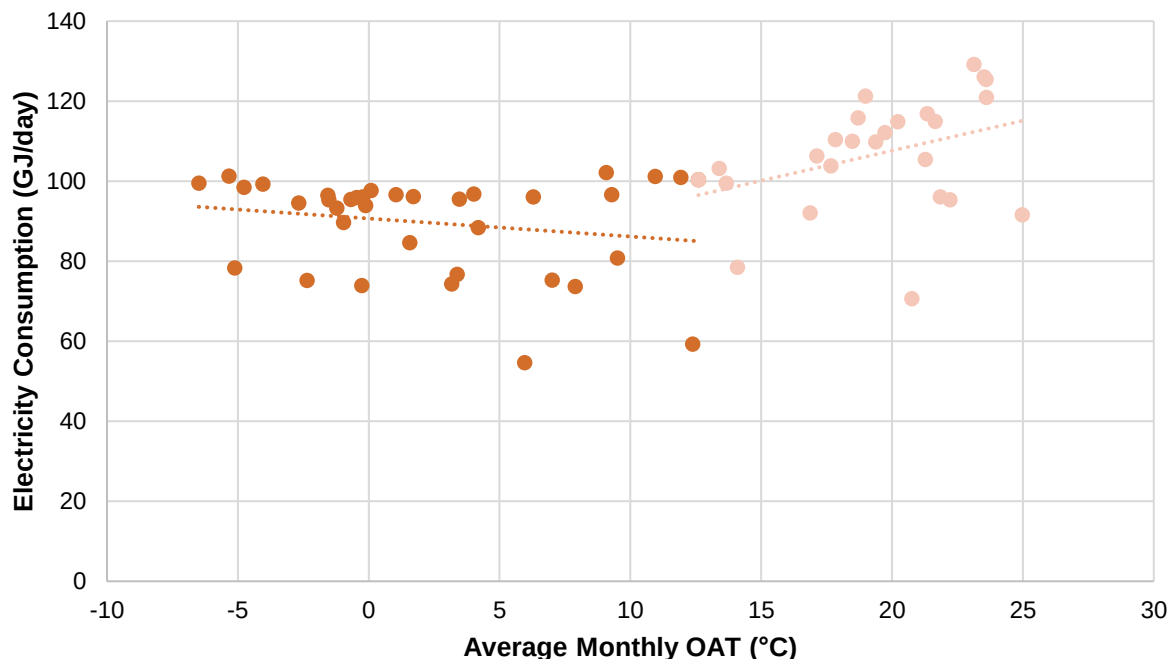


Figure 11: Electricity Consumption vs. Average Monthly OAT.

The electricity consumption is plotted against cooling degree days with a base temperature of 13°C in Figure 12. With an R-squared value of 0.315, electricity consumption shows a weak correlation to cooling degree days.

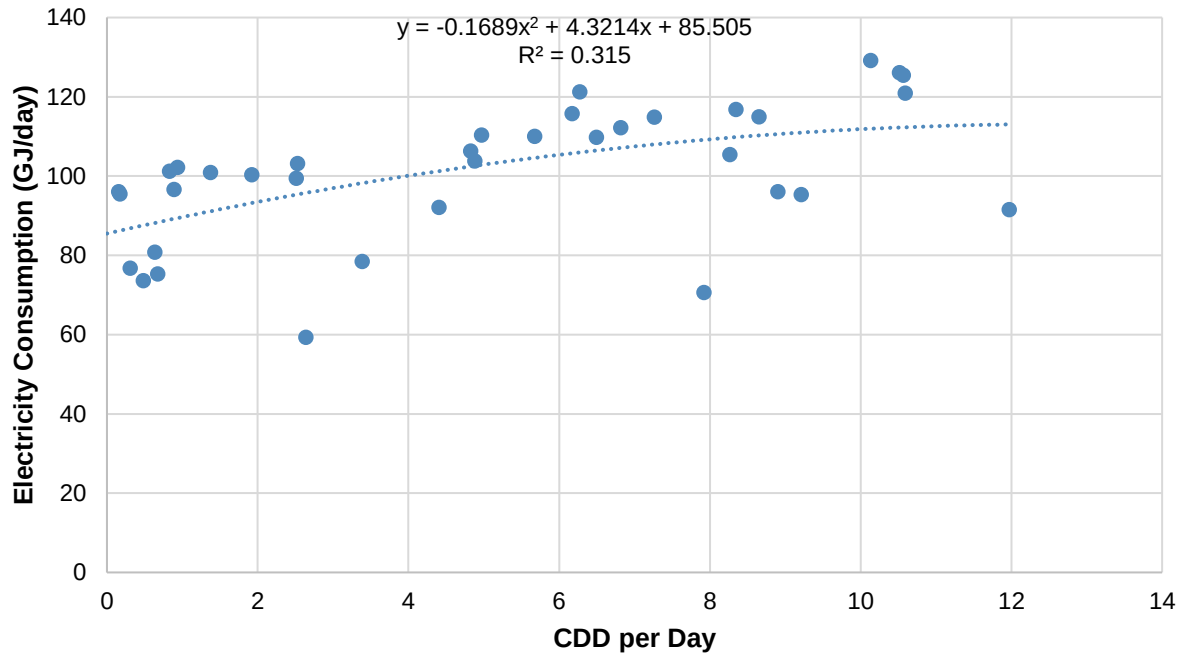


Figure 12: Electricity Consumption vs. Cooling Degree Days

Natural Gas

The weather sensitivity of the building's natural gas consumption was examined by plotting the average outdoor air temperature against daily consumption for each month in Figure 13.

There appears to be two distinct trends – one weather dependent and one weather independent. An approximate trend line was overlaid on the data to help determine the inflection point, which is about 10°C. Below the inflection point, gas consumption is dominated by space heating and humidification loads; while above the inflection, gas use is relatively independent of weather and is used for domestic hot water and cooking. In a few months, such as December 2020, January 2021, February 2021, and April 2020, gas consumption did not follow patterns; the cause is unclear.

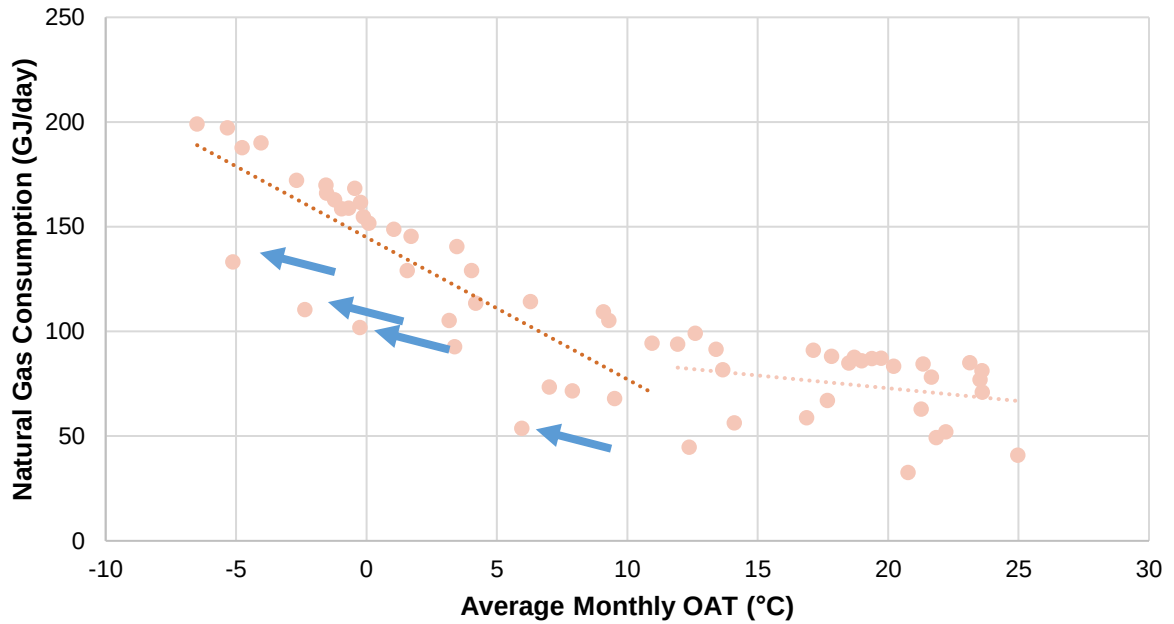


Figure 13: Average Outdoor Air Temperature vs. Natural Gas Consumption

Gas consumption is plotted against heating degree days with a base temperature of 10.0°C in Figure 14. With an R-squared value of 0.78, gas use shows a moderate relationship with heating degree days. Moderate correlation indicates that weather normalized energy use estimates should be moderately dependable.

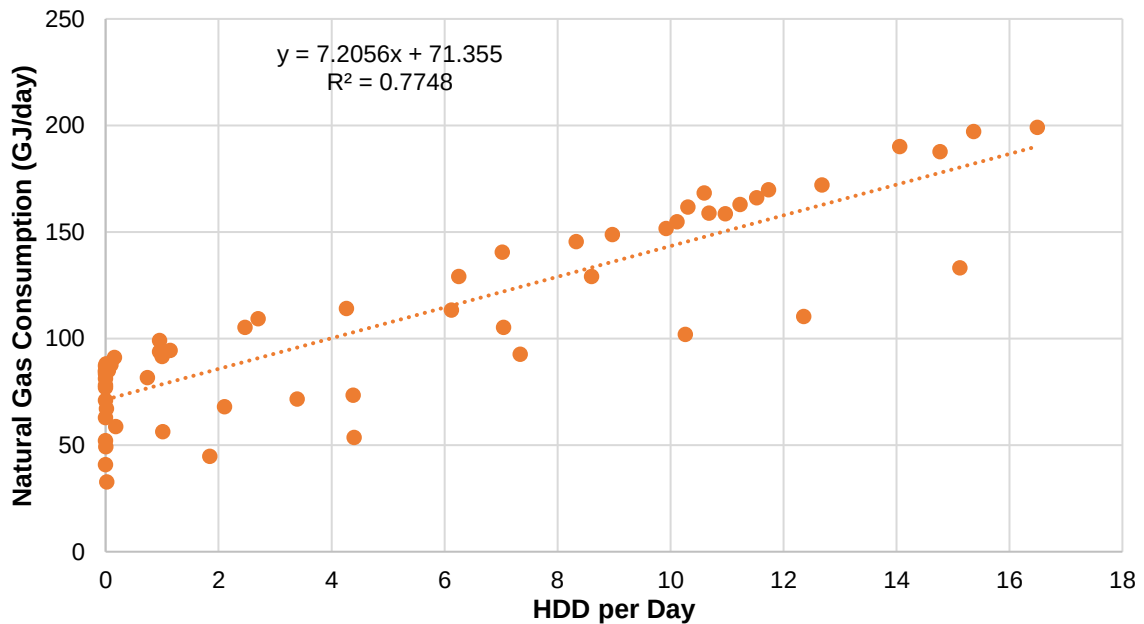


Figure 14: Heating Degree Days vs. Natural Gas Consumption

APPENDIX E: Energy Efficiency Measure Details

5.3.1 Replace Lavatory Aerators with Low Flow

Current Condition

Washrooms are equipped with lavatory faucets with standard flow aerators. Based on a survey of plumbing fixtures, the approximate quantities are summarized below.

Flow (Lpm)	5.7
Quantity	333

Measure Description

Replace higher flow lavatory aerators with 1.9 Lpm models.

Implementation Considerations

Retain contractor to replace aerators as required. Dispose of spare high flow aerators in stock room.

Calculation of Energy Savings

Lavatory water savings have been calculated using fixture flow rates and assuming: 3 daily use per occupant; 60 seconds per use; 250 (60% of full capacity of guest and staffs) occupants; and 365 workdays.

Domestic hot water savings were calculated using 30% of the lavatory flow and a heater efficiency of 80% as follows:

$$DHW \text{ Energy } kJ = \frac{DHW \text{ m}^3 \times \text{density } \frac{kg}{m^3} \times \text{Specific Heat } \frac{kJ}{kg \cdot K} \times \text{Delta } T \text{ } 45^{\circ}C}{\text{efficiency}}$$

Calculation of Maintenance Savings

None identified.

Ancillary Benefits

None identified.

Project Cost Summary

Make an allowance for \$80 per aerator for parts and labour for replacement.

5.3.1 Replace Lavatory Aerators with Low Flow

Annual Savings						Initial Costs and Payback			
Electricity (kWh)	Electricity Peak (kW)	Natural Gas (m ³)	Water (m ³)	Utility Cost (\$)	Maintenance (\$/year)	Opinion of Cost (\$)*	Incentive (\$)**	Simple Pay-back (years)	Simple ROI
-	-	4,739	1,040	\$5,884	-	\$42,000	-	7.2	14%

* Includes HST. ** Estimate of one time incentive payment, contingent on third party approval; programs may change without notice.

5.3.2 Replace Shower Heads with Low Flow

Current Condition

Guest rooms and other showers are equipped with standard flow shower heads. Based on a survey of plumbing fixtures, the approximate flow rates and quantities are summarized below.

Flow (Lpm)	9.5
Quantity	270

Measure Description

Replace higher flow shower heads with 5.7 Lpm models.

Implementation Considerations

Retain contractor to replace shower head as required. Dispose of spare high flow aerators in stock room.

Calculation of Energy Savings

Shower heads water savings have been calculated using fixture flow rates and assuming: 1 daily use per occupant; 10 minutes per use 250 (60% of full capacity of guest and staffs) occupants; and 365 workdays.

Domestic hot water savings were calculated using 50% of the sink flow and a heater efficiency of 80% as follows:

$$DHW \text{ Energy } kJ = \frac{DHW \text{ m}^3 \times \text{density } \frac{kg}{m^3} \times \text{Specific Heat } \frac{kJ}{kg \cdot K} \times \Delta T \text{ } 45^{\circ}C}{\text{efficiency}}$$

Calculation of Maintenance Savings

None identified.

Ancillary Benefits

None identified.

Project Cost Summary

Make an allowance for \$80 per shower head for parts and labour for replacement.

5.3.2 Replace Shower Heads with Low Flow

Annual Savings						Initial Costs and Payback			
Electricity (kWh)	Electricity Peak (kW)	Natural Gas (m ³)	Water (m ³)	Utility Cost (\$)	Maintenance (\$/year)	Opinion of Cost (\$)*	Incentive (\$)**	Simple Pay-back (years)	Simple ROI
-	-	18,881	3467	\$20,189	-	\$34,000	-	1.69	59%

* Includes HST. ** Estimate of one time incentive payment, contingent on third party approval; programs may change without notice.

5.4.1 Replace Urinals

Current Condition

Washrooms are equipped with standard flow urinals. Based on a survey of plumbing fixtures, the approximated quantities are summarized below.

Flow (Lpf)	3.8
Quantity	15

Measure Description

Replace the standard flow urinals with 1.9 LPF models.

Implementation Considerations

Retain a contractor to replace the urinals.

Calculation of Energy Savings

Urinal replacement water savings have been calculated using fixture flow rates and assuming: 2 daily use per male occupant and 365 workdays.

Calculation of Maintenance Savings

None identified.

Ancillary Benefits

None identified.

Project Cost Summary

Provide \$1,000 per urinal for the purchase of new low flow urinal and labour costs associated.

Annual Savings						Initial Costs and Payback			
Electricity (kWh)	Electricity Peak (kW)	Natural Gas (m ³)	Water (m ³)	Utility Cost (\$)	Maintenance (\$/year)	Opinion of Cost (\$)*	Incentive (\$) **	Simple Pay-back (years)	Simple ROI
-	-	-	34.7	158	-	19,000	-	119	1%

* Includes HST. ** Estimate of one time incentive payment, contingent on third party approval; programs may change without notice.

5.4.2 Replace Water Closet with Ultra Low Flow Fixtures

Current Condition									
Washrooms are equipped with standard flow water closets. Based on a survey of plumbing fixtures, the approximated quantities are summarized below.									
		Flow (Lpm)		6.0					
		Quantity		313					
Measure Description									
Replace the standard flow water closets with 4.8 LPF models.									
Implementation Considerations									
Retain a contractor to replace the water closets.									
Calculation of Energy Savings									
Urinal replacement water savings have been calculated using fixture flow rates and assuming: 3 daily use per occupant and 365 workdays.									
Calculation of Maintenance Savings									
None identified.									
Ancillary Benefits									
None identified.									
Project Cost Summary									
Provide \$1,500 per urinal for the purchase of new low flow water closet and labour costs associated.									
Annual Savings						Initial Costs and Payback			
Electricity (kWh)	Electricity Peak (kW)	Natural Gas (m³)	Water (m³)	Utility Cost (\$)	Maintenance (\$/year)	Opinion of Cost (\$)*	Incentive (\$) **	Simple Pay-back (years)	Simple ROI
-	-	-	328	338	-	396,000	-	-	-

* Includes HST. ** Estimate of one time incentive payment, contingent on third party approval; programs may change without notice.

5.4.3 Install Occupancy Motion Sensors in the Stairwell

Current Condition

T8 LED fluorescent luminaires are used to illuminate the stairwells throughout the facility on a 24/7 schedule.

Measure Description

Re-lamp the stairwell fluorescent luminaires in bulk with LED equivalents with inbuilt occupancy sensors.

Implementation Considerations

Retain an electrician to re-lamp the existing fluorescent luminaires with LED equivalents.

For a list of replacements eligible for SaveONenergy incentives, refer to their website. Coupons for incentives are also available on their website and at some retailers.

Calculation of Energy Savings

These luminaires are estimated to operate 24 hours/day on average throughout the year. The input power and energy use of the existing fixtures are summarized below.

Fixture	Qty.	Input W	Total kW	Hours/y	kWh/y
Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB LED	98	32	0	8,760	28,837

The input power and energy use of LED replacement lamps are summarized below.

Fixture	Qty.	Input W	Total kW	Hours/y	kWh/y
Type F2 48" T8 Wrap Surface Mounted Fluorescent 2LB LED + Occ Sensor	98	32	0	4,380	14,418

The proposed LED occupancy motion sensors is estimated to save 14,418 kWh per year, with an energy cost savings of approximately \$1,649 per year.

Ancillary Benefits

Fluorescent lamps contain mercury, which can contaminate the environment. Replacement with LED eliminates this source of mercury.

Project Cost Summary

The replacement cost for each luminaire is estimated at \$100 each and labour at \$120/hour for a technician. It is assumed the technician could install 1 fixture per hour.

5.4.3 Install Occupancy Motion Sensors in the Stairwell

Annual Savings						Initial Costs and Payback			
Electricity (kWh)	Electricity Peak (kW)	Natural Gas (m ³)	Water (m ³)	Utility Cost (\$)	Maintenance (\$/year)	Opinion of Cost (\$)*	Estimated Incentive (\$) **	Simple Pay-back (years)	Simple ROI
14,418	-	-	-	\$1,649	\$429	\$8,600	\$0	4.1	24%

* Includes HST. ** Estimate of one time incentive payment, contingent on third party approval; programs may change without notice.

5.5.1 Replace Gas Fired Hot Water Boiler with High Efficiency Condensing Boilers

Current Condition

The building is equipped with four low pressure gas hot water boilers which serves the FCUs, AHUs and MAUs.

Measure Description

Replace the existing low pressure hot water boilers with high efficiency condensing boilers at the end of useful service life.

Implementation Considerations

Calculation of Energy Savings

Based on the end use breakdown, 26,654 GJ of natural gas is being used for space heating, pool and other space heating purposes. Since the existing hot water boilers have an efficiency of about 85% there is some energy being dissipated due to equipment inefficiencies. However, with a high efficient condensing boiler, it will operate almost at 96% efficiency.

Calculation of Maintenance Savings

None identified.

Ancillary Benefits

Condensing boiler will also reduce the GHG emissions of the building.

Project Cost Summary

We have made an allowance for:

- Four condensing boilers price as per RSMeans 2017: $\$376,000 \times 4 = \$1,504,000$
- Engineering support: \$20,000

Annual Savings							Initial Costs and Payback			
Electricity (kWh)	Electricity Peak (kW)	Natural Gas (m ³)	Water (m ³)	Utility Cost (\$)	Maintenance (\$/year)	GHG Emissions (kg CO ₂)	Opinion of Cost (\$)*	Incentive (\$)**	Simple Pay-back (years)	Simple ROI
-	-	77,879	-	97,703	-	153,712	\$1,504,000	-	-15	-7

* Includes HST. ** Estimate of one time incentive payment, contingent on third party approval; programs may change without notice.

5.5.2 Install HRVs on Air Handling Units ACS-1, 4 5, and 6

Current Condition										
The building is equipped with 17 AHUs; among them, units 7, 9, 12, and 13 are equipped with heat recovery wheels. Additionally, two MAUs have wheels with heat recovery. However, several other units could incorporate heat recovery (HRVs) devices. The addition of heat recovery ventilation units is advised for the ACS-1, 4, 5, and 6 units.										
Measure Description										
Install HRVs with AHUs ACS-1, 4, 5 and 6 to save exhaust energy from the conditioned spaces.										
Implementation Considerations										
Calculation of Energy Savings										
With similar indoor temperature set points and a building's envelope, 2,388GJ of natural gas can be saved based on hourly weather data and a calculation on an Excel spreadsheet's assessment of hourly heat loss.										
Calculation of Maintenance Savings										
None identified.										
Ancillary Benefits										
HRV installation will also reduce the GHG emissions of the building.										
Project Cost Summary										
We have made an allowance for:										
- Four HRV system installation with existing AHUs price: \$90,000 - Engineering support: \$10,000										
Annual Savings							Initial Costs and Payback			
Electricity (kWh)	Electricity Peak (kW)	Natural Gas (m³)	Water (m³)	Utility Cost (\$)	Maintenance (\$/year)	GHG Emissions (kg CO₂)	Opinion of Cost (\$)*	Incentive (\$)***	Simple Pay-back (years)	Simple ROI
-	-	60,855	-	14,299	-	120,171	\$100,000	-	7	14%

* Includes HST. ** Estimate of one time incentive payment, contingent on third party approval; programs may change without notice.

5.5.4 Install HRVs System with Kitchen Ecology Units KEU-1,2,3,4 & 5.

Current Condition										
The building is equipped with five kitchen ecology units with a combined exhaust of 21,703 cfm. According to ASHRAE 90.1, it is advised to install HRVs into every kitchen ecology unit in order to create a healthy kitchen environment.										
Measure Description										
Install HRVs with kitchen ecology units KUE-1,2,3,4 & 5 to save exhaust energy from the conditioned spaces.										
Implementation Considerations										
Calculation of Energy Savings										
With similar indoor temperature set points and a building's envelope, 2,840GJ of natural gas can be saved based on hourly weather data and a calculation on an Excel spreadsheet's assessment of hourly heat loss.										
Calculation of Maintenance Savings										
None identified.										
Ancillary Benefits										
HRV installation will also reduce the GHG emissions of the building.										
Project Cost Summary										
We have made an allowance for:										
- Four HRV system installation with existing AHUs price: \$85,000 - Engineering support: \$10,000										
Annual Savings							Initial Costs and Payback			
Electricity (kWh)	Electricity Peak (kW)	Natural Gas (m³)	Water (m³)	Utility Cost (\$)	Maintenance (\$/year)	GHG Emissions (kg CO₂)	Opinion of Cost (\$)*	Incentive (\$)**	Simple Pay-back (years)	Simple ROI
-	-	72,417	-	17,008	-	124,933	\$95,000	-	6.0	18%

* Includes HST. ** Estimate of one time incentive payment, contingent on third party approval; programs may change without notice.